



In depth: Principles of interaction, methods of interaction in visualization

Visualization of Information

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#	Content	Comment
1	High level overview of the steps of ML and discern when and why visualizations can be used to improve the process	Theory
2	Introduction: Why/What/How visualization pipe, semiology, variables, and taxonomy	Theory
3	Principles of human perception, principles of information processing and perception, saliency, etc.	Theory
4	Data sources and processes of data processing (pre-processing – data transformations)	Theory
5	Visualization methods for temporal data (time-series) and multi-dimensional data (multi variate)	Theory + Hands On
6	Interactive Visualizations with Shiny R (as an example) and other platforms (PowerBI) – how to use gen AI tools to generate visualizations	Theory + Hands On
7	In depth: Principles of interaction, methods of interaction in visualization	Theory + Hands On
8	Lie and truth in information presentation and Dashboards	Theory
9	Visualization of statistical information (e.g. variance, etc.) and visual analytics in machine learning	Theory + Hands On
10	Visualization of text and documents Visualization methods for trees, graphs, and networks	Theory + Hands On
11	Comparison and Evaluation of Visualizations + Story Telling	Theory + Hands On
12	Presentations	
13		



Agenda

1. Short review from last session – questions and answers
2. Why?
3. What Is Perception?
4. Perceptual Processing
5. Resource Model of Human Information Processing
6. Metrics
7. Cognition

Why Interactions Matter in Visualizations

1. Promote Active Exploration: Users engage more deeply by manipulating views, not just observing them.
2. Supports Curiosity and Information Foraging: Dynamic exploration can reveal patterns and anomalies that static views may miss.
3. Adapt to User Goals: Interactions let users tailor the visualization to their specific questions and tasks.
4. Support Iterative Thinking: Users refine their understanding through trial, error, and continuous refinement.
5. Enhance Trust and Engagement: Interactive systems feel responsive and user-centered, boosting confidence in the data.

Many options

1. R
2. Python
3. PowerBI
4. Tableau – Sales Force
5. Excel
6. Cognos - IBM
7. AI
8. Business Objects – SAP
9. ...

What has changed with AI?

- AI tools generate charts/dashboards from plain natural language prompts.
- Less need to memorize Python/R syntax.
- AI auto-cleans data and fixes code errors.
- MCP-connected AI can query data sources directly.
- Focus is shifting from coding skills to analytical thinking.

MCP

- MCP stands for Model Context Protocol.
- Standard that lets AI models connect to external tools and data sources safely and dynamically.
- Allows AI tools like Claude Code or OpenAI agents to:
 - access databases
 - read files
 - call APIs
 - use GitHub
 - query Whatsapp/Slack/Jira
 - interact with development tools automatically.

PowerBI

Building blocks of Power BI – low code

1. Create a semantic model

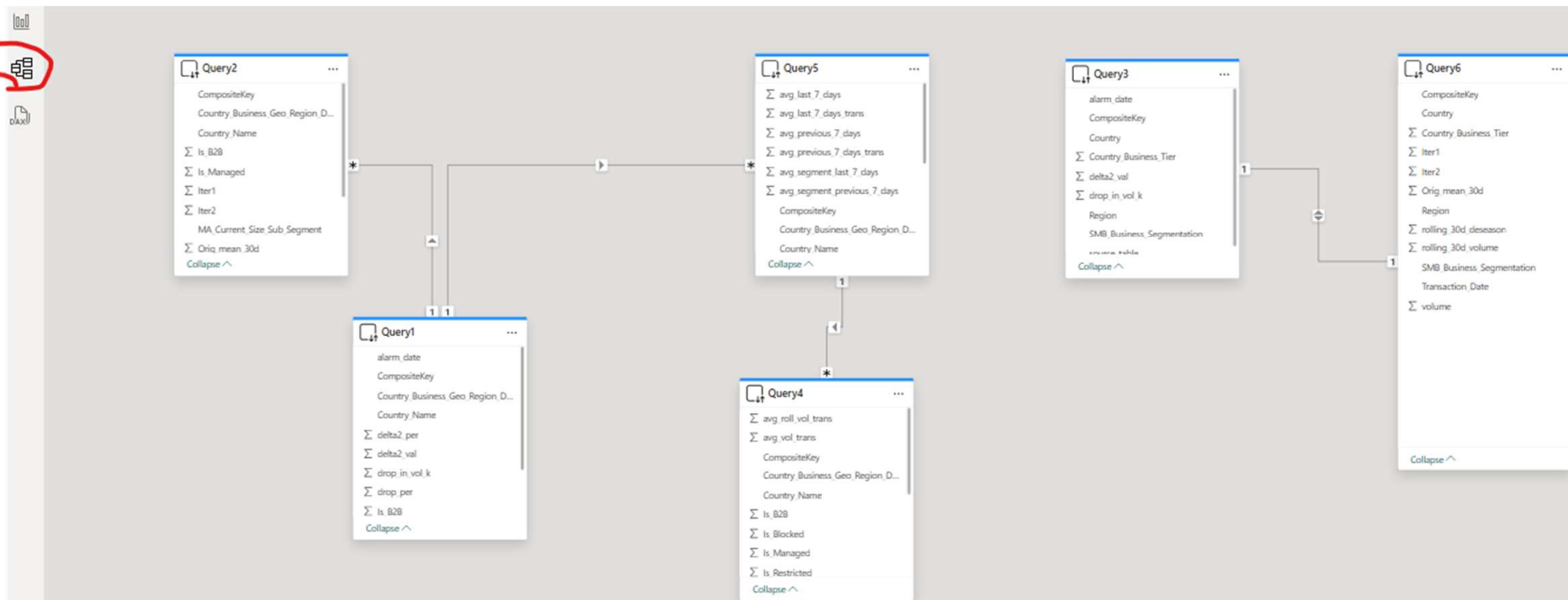
- consists of all connected data, transformations, relationships, and calculations.

2. Create visualizations:

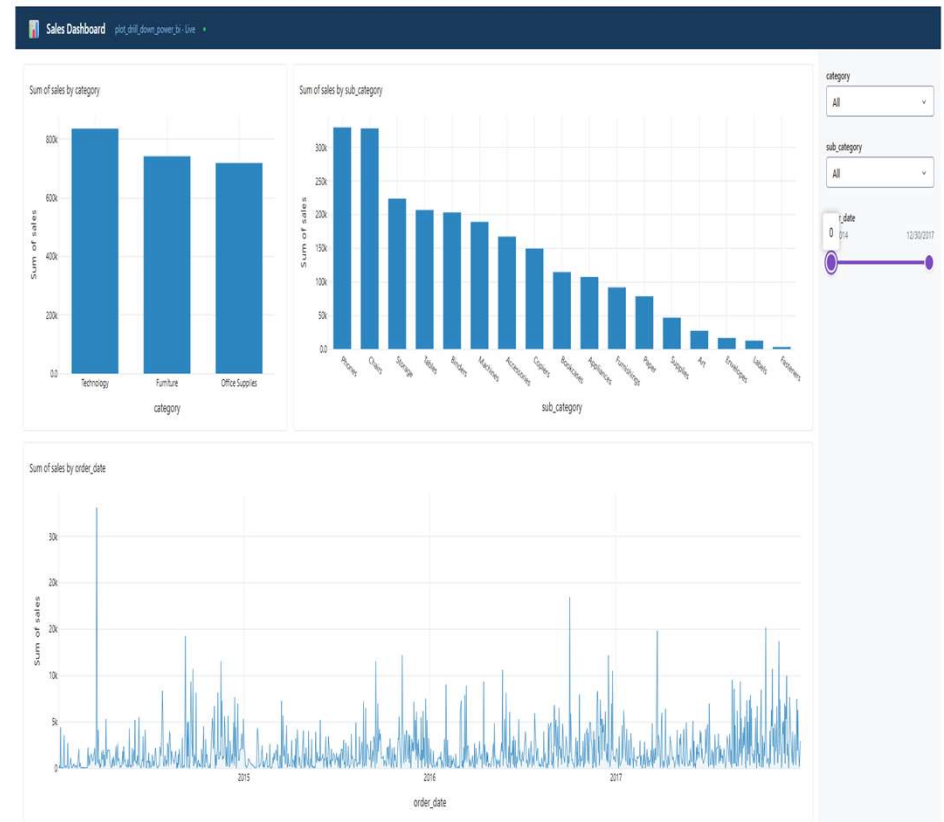
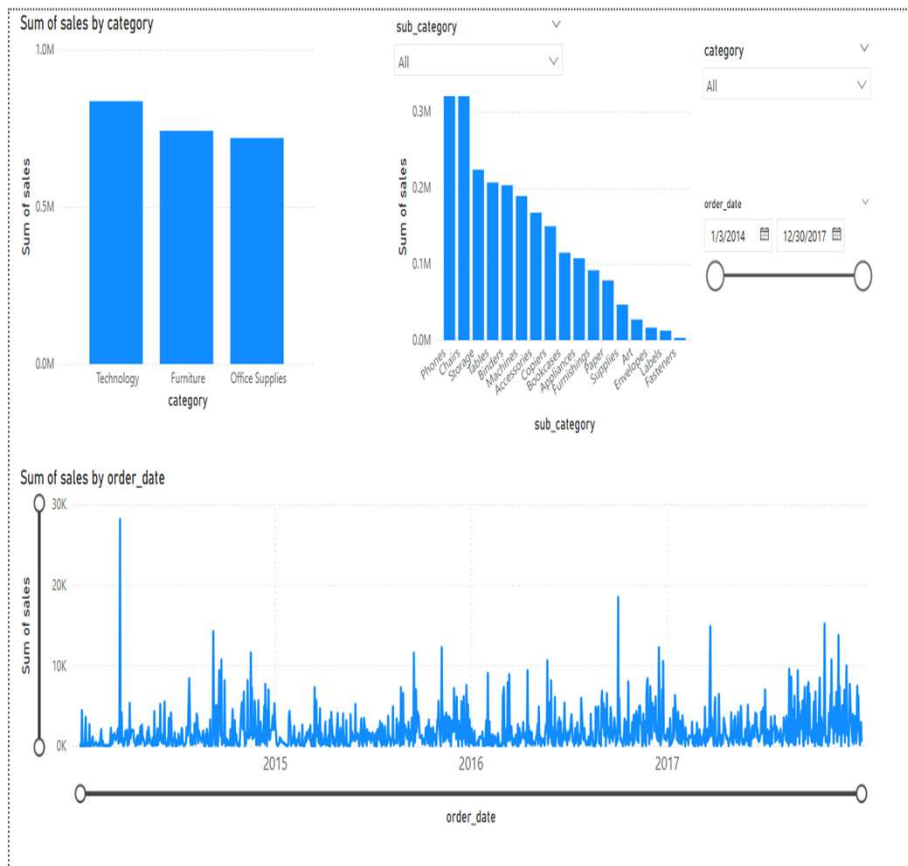
- add them to the canvas for a report page.
- drag and drop
- Link/interactivity between visuals

3. Publish report to Power BI service.

PowerBI – semantic model



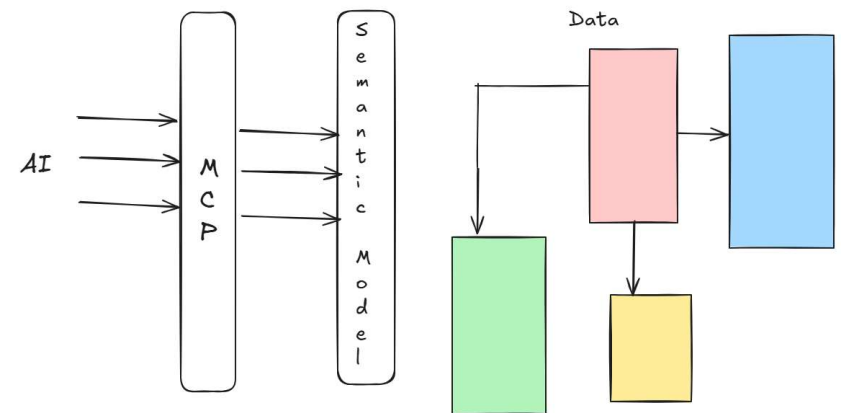
PowerBI and MCP to build a dashboard using Python



07/17/2020

Why AI Needs Access to the Semantic Model

1. Trust needs Data Governance
2. When AI generates code based on the semantic model, every developer on the team gets code that references the same single source of truth — preventing the proliferation of inconsistent KPI definitions across dashboards
3. Without this, different students/developers may build dashboards where "Total Sales" means different things — a classic data governance failure
4. AI generating code without semantic model access is like a new employee writing reports without reading the data dictionary — the output may look correct but contain fundamental logical errors that only surface when business decisions are made on wrong numbers



Interaction Support

- Simple Information seeking mantra proposed by Shneiderman (1996) should guide the users when exploring the data visually:
 - ✓ Overview first,
 - ✓ zoom and filter,
 - ✓ then details-on-demand
- More complete (Keim et al. (2006, p. 6)):
 - ✓ Analyze First -
 - ✓ Show the Important -
 - ✓ Zoom, Filter and Analyze Further -
 - ✓ Details on Demand

Interaction Support

- Visual exploration and analysis is not a one-way street where data are transformed into images.
- High degree of interactivity and advanced interaction techniques for working with data are important.
- Interaction helps users in:
 - ✓ understanding the visual mapping
 - ✓ realizing the effect of visualization parameters
 - ✓ exposing hidden patterns
 - ✓ trusting the data
- Interaction also provokes curiosity – users want to get their hands on their data – which is particularly useful when exploring unknown data.

Interaction Support

- **Interactive exploration is iterative:**
Users shift attention across different parts of the data, explore from various perspectives, and investigate multiple questions.
- **Start with an overview:**
Use appropriate data abstraction and tailored visual representations to help users identify areas for deeper analysis (Shneiderman, 1996).

Interaction Support

- **Explore related or alternative views:**

Users may navigate to related or similar data or return to the overview to explore from a new perspective or address a different question.

- **Build mental models through interaction:**

By moving from one focus to another—whether it's a data subset, aspect, or analysis task—users develop understanding and gain insights.

Reminder:

Interaction Support – Interaction operators

1. Select – Mark something as interesting
2. Explore – Show me something else (navigate through the data)
3. Reconfigure – Show me a different arrangement of the data
4. Encode – Show me a different representation
5. Abstract/Elaborate – Show me more or less detail
6. Filter – Show me something conditionally
7. Connect – Show me related items
8. Undo/Redo – Let me go to where I have already been
9. Change configuration – Let me adjust the interface

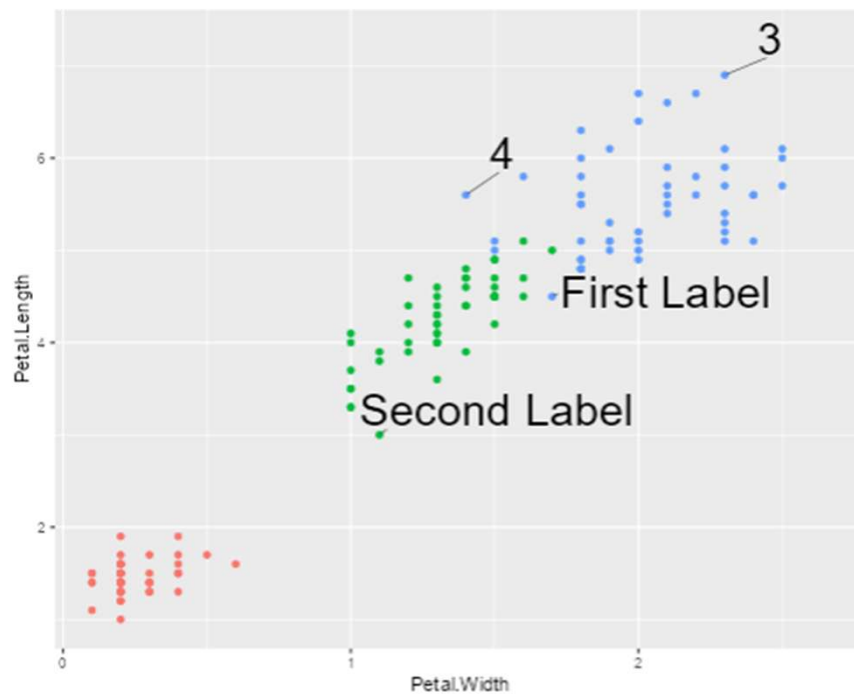
Interaction Operators - selection

- Select operation allows users to mark and visually highlight interesting findings or important analysis results in a visual representation.
- Users can isolate a subset of display components through selection and apply other operations like highlighting, deleting, masking, or focusing.

Interaction Operators - selection

- Selection can:
 - replace or supplement previous selections, and the granularity of selection can vary from the smallest addressable component to a broader region.
 - be performed through clicking, painting over entities, bounding boxes, lassoes, or indirect methods based on user-specified constraints.
 - help users keep track of items of interest in large datasets or with changes in representations.
 - be often used as a preceding action to subsequent operations, allowing users to see the location of items of interest in new arrangements.
 - be typically combined with other techniques to enhance user exploration and discovery.

Interaction Operators – selection example

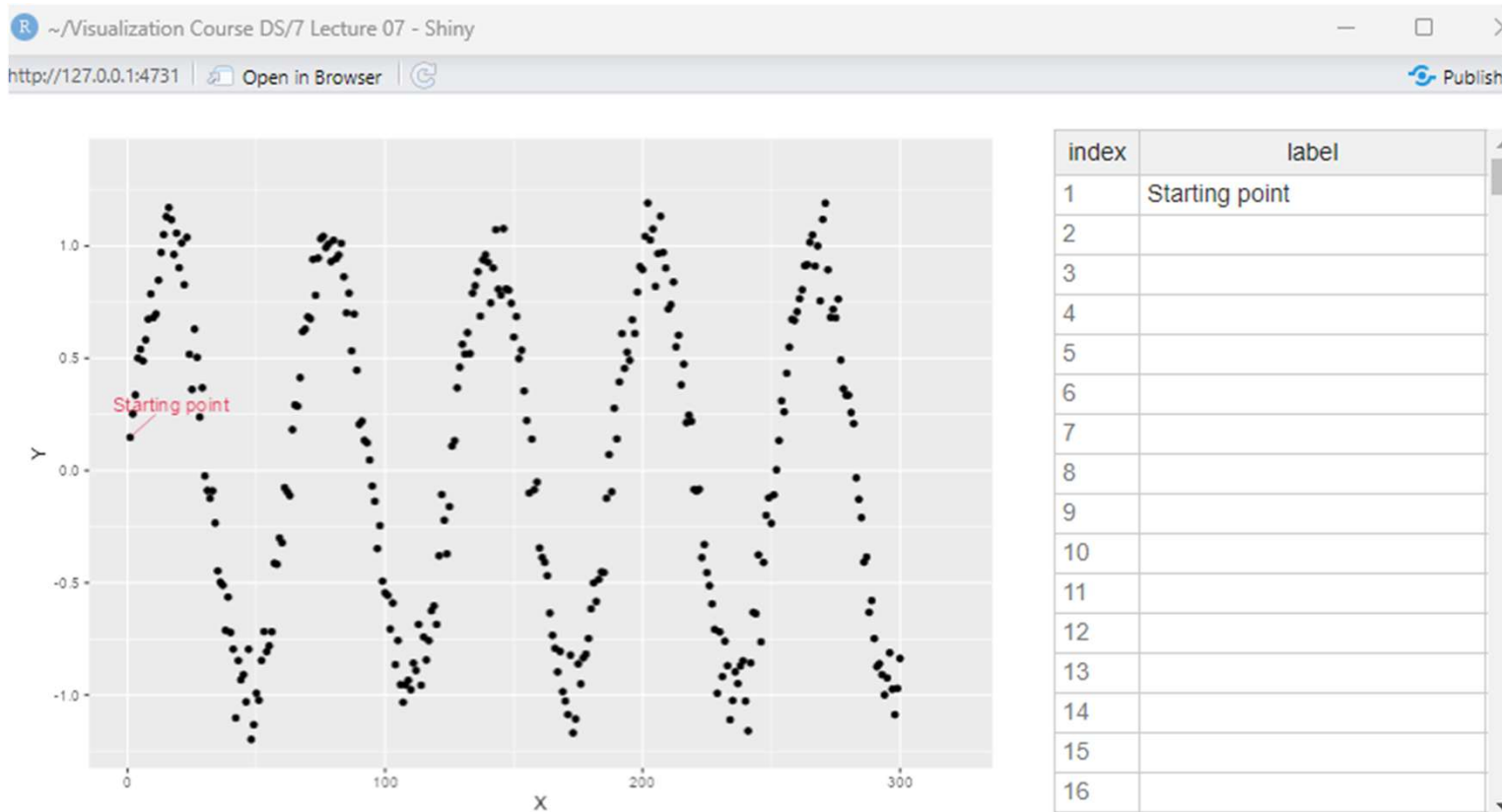


Using Shiny R

Source: <https://www.r-bloggers.com/2022/08/labelling-the-points-of-a-ggplot-with-shiny/>

Script: Labeling

Interaction Operators – selection example



Script: Labeling_1

Interaction Operators – selection example

(A) Select



Interaction Operators - exploration

- Explore operators involve searching for and examining subsets of data, determining the view orientation, and adjusting the level of detail.
- Exploration can be based on visual patterns or the need for more detailed analysis.
- It involves specifying camera location, viewing direction, frustum shape and size, and level of detail.
- Exploration can work with absolute or relative coordinates and can be driven by the user or automated.

Interaction Operators - exploration

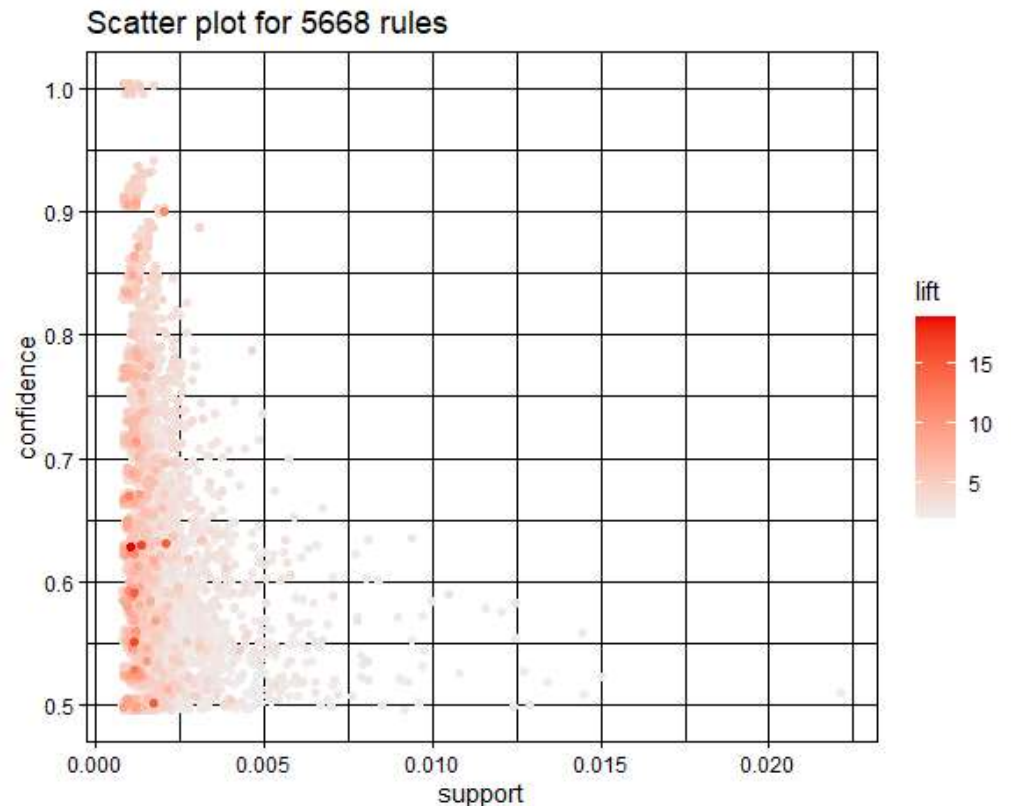
- Explore interaction techniques allow users to examine different subsets of data cases.
 - Panning is a common exploration used by moving the viewport to visualize other areas of the content, which can usually be performed using the scroll bars.
 - Direct-Walk enables smooth movement of the viewing focus from one position to another, often achieved through mouse points or direct manipulation methods. Allows users to smoothly change the focus of the view.
 - Automated exploration techniques include the grand tour and projection pursuit. The visualization of complex multivariate time-oriented data requires interactive exploration of different time ranges and data variables.

Zooming ,Panning, Brushing

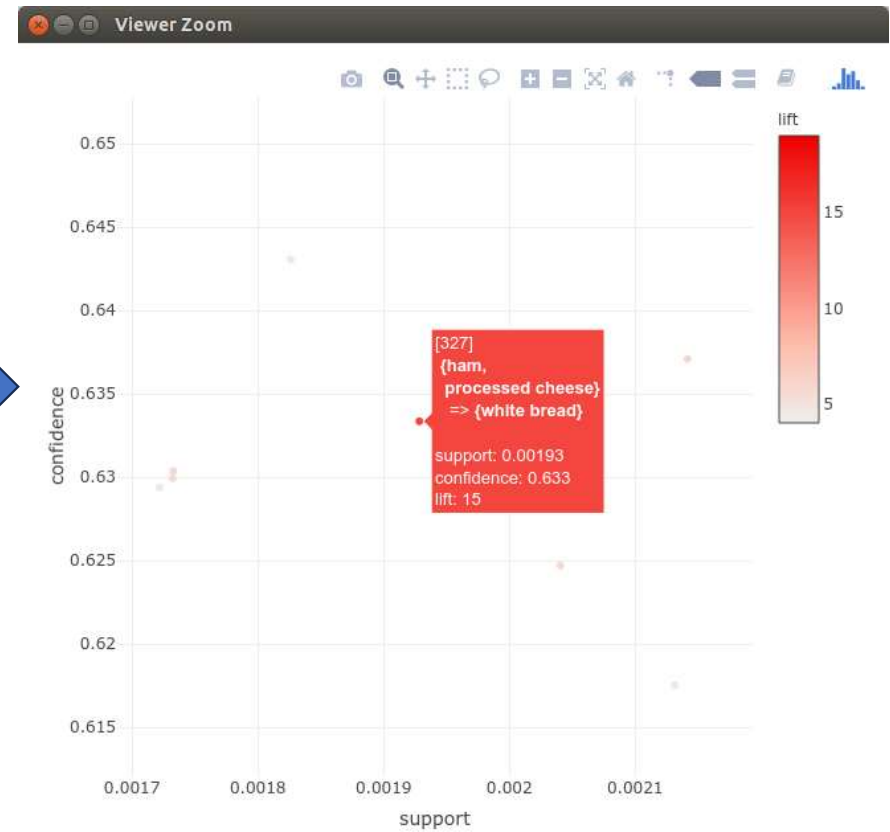
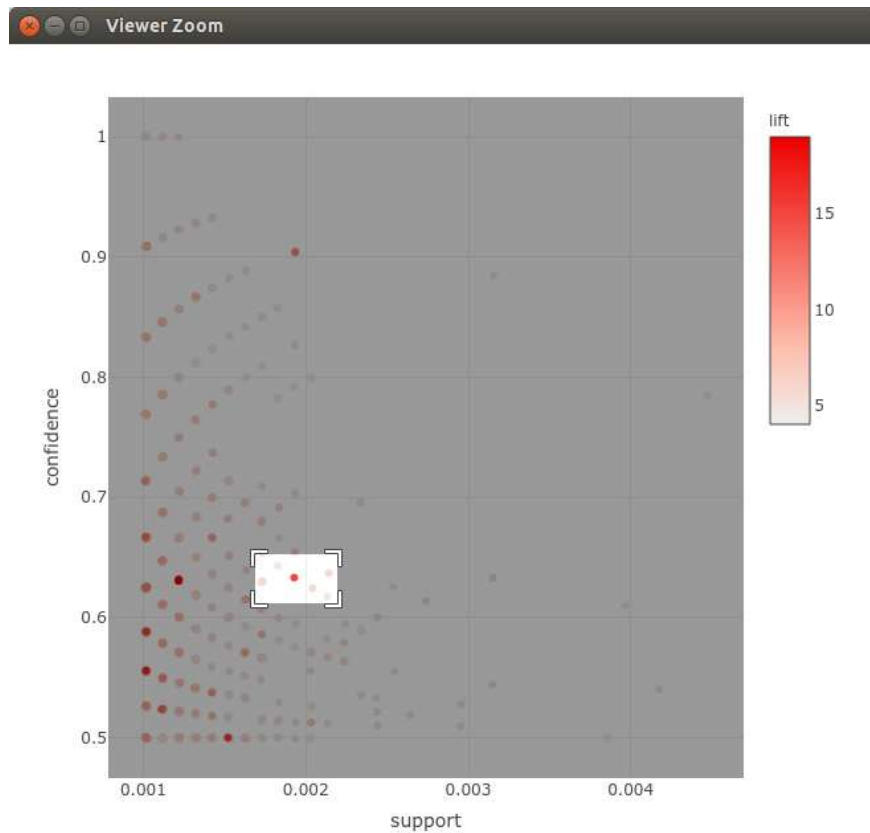
- **Zooming:** Zooming changes the scale of the view to show more or less detail.
 - Explore data at different levels of granularity.
 - Focus on a smaller region for detailed inspection.
 - Reveal patterns hidden at certain scales.
 - Example: Zooming into a section of a time series plot to examine a local anomaly.
- **Panning:** Panning moves the visible window across the dataset without changing the scale.
 - Navigate across large datasets.
 - Maintain zoom level while exploring different parts of the data.
 - Example: After zooming into a time period, you pan left or right to examine adjacent time windows
- **Brushing:** Brushing lets the user select a subset of data points directly on the visualization (usually by dragging a rectangle or lasso).
 - Highlight or filter data points for further analysis.
 - Link selections across multiple views (e.g., brushing in a scatterplot highlights rows in a table).

Examples using package arulesViz from R

- The package arulesViz (Hahsler 2017) is part of the arules package ecosystem for handling and mining association rules (Hahsler et al. 2011).
- small demo data set called “Groceries” which is included in arules



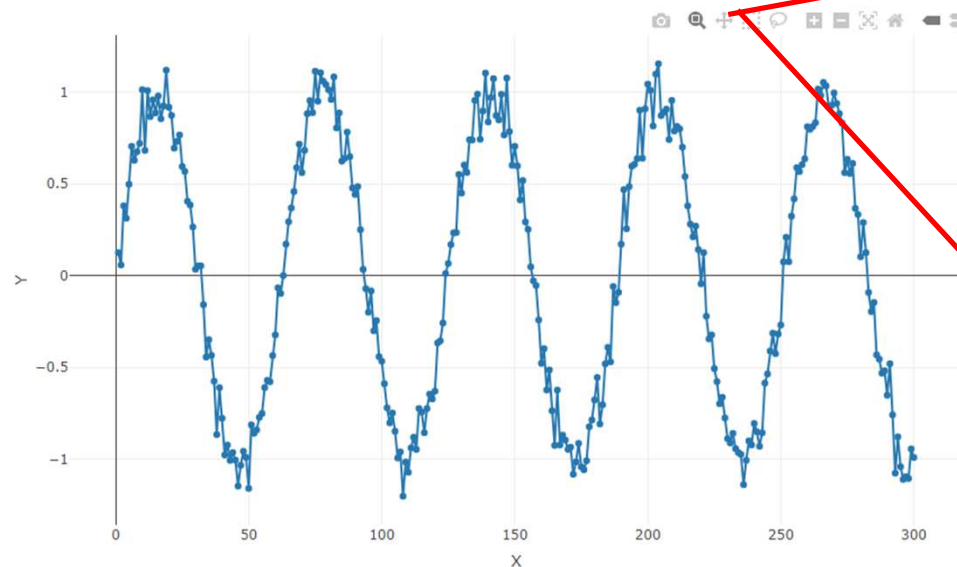
Examples using package arulesViz from R



Examples of Panning



Pan and Zoom Example (Plotly)

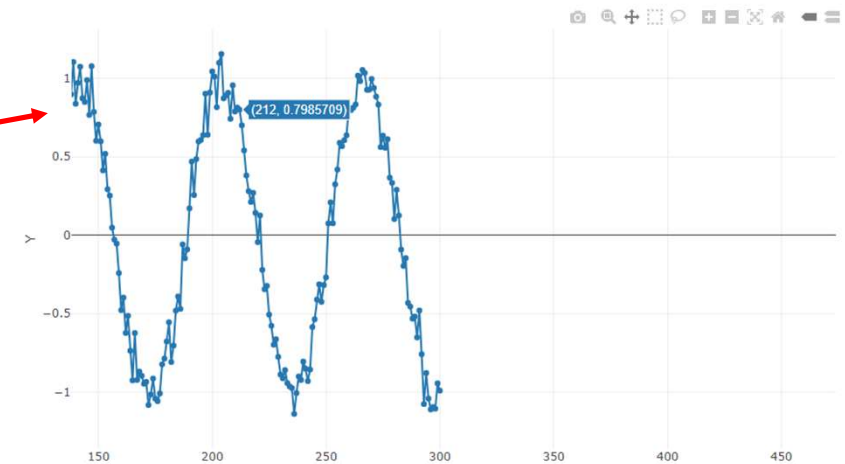


Use mouse scroll to zoom. Use click and drag to pan. Use the toolbar to zoom or reset.

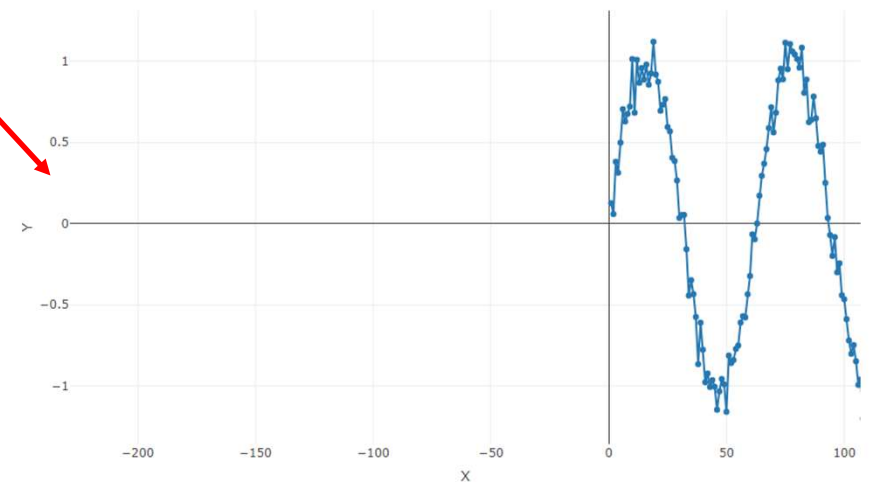
Using plotly: panning script

6/13/2026

Pan and Zoom Example (Plotly)



Pan and Zoom Example (Plotly)

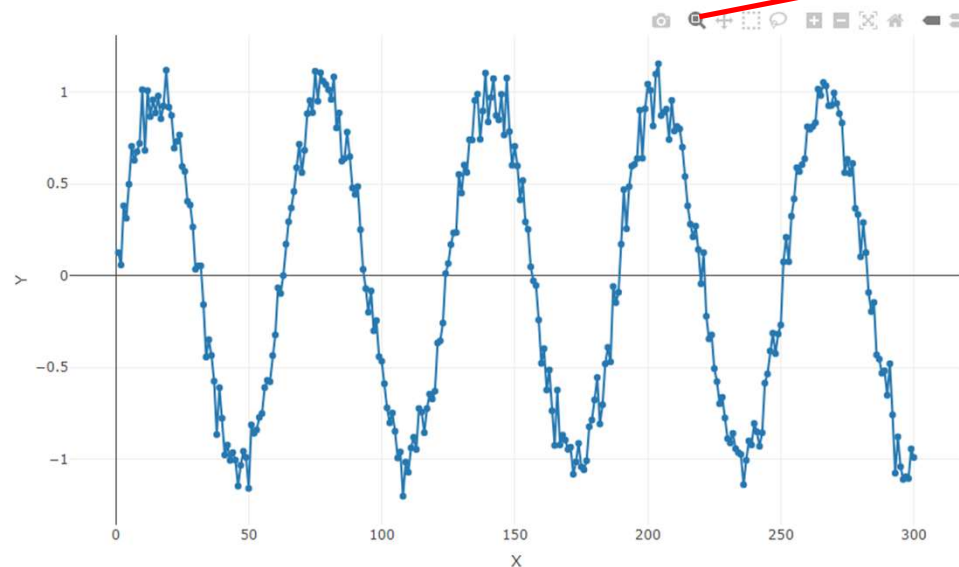


Use mouse scroll to zoom. Use click and drag to pan. Use the toolbar to zoom or reset.

Examples of Zooming

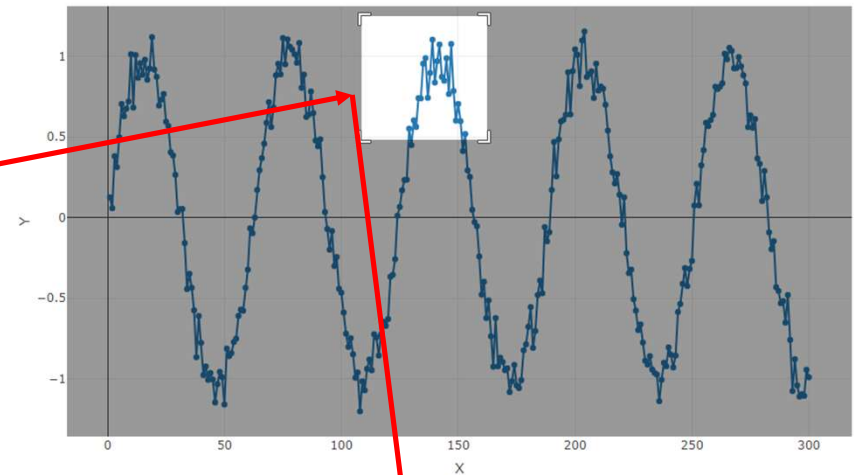
~/Visualization Course DS/7 Lecture 07 - Shiny
http://127.0.0.1:4731 | Open in Browser | Publish

Pan and Zoom Example (Plotly)



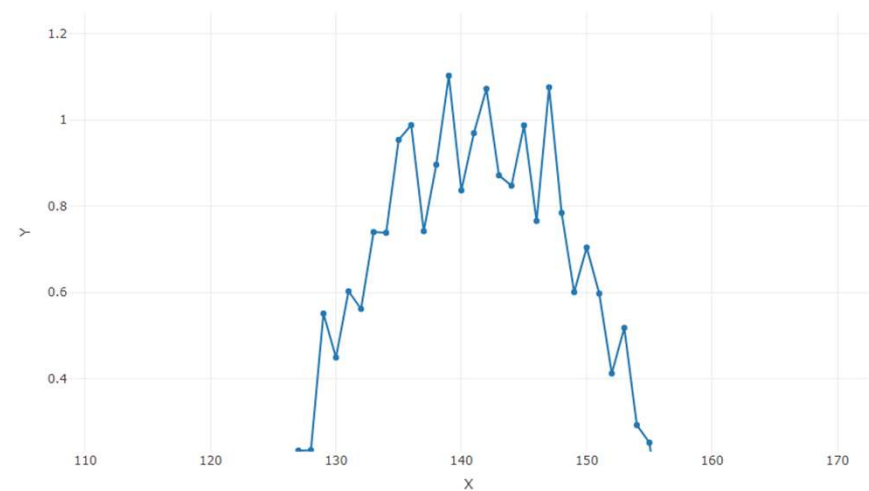
Use mouse scroll to zoom. Use click and drag to pan. Use the toolbar to zoom or reset.

Pan and Zoom Example (Plotly)



Use mouse scroll to zoom. Use click and drag to pan. Use the toolbar to zoom or reset.

Pan and Zoom Example (Plotly)

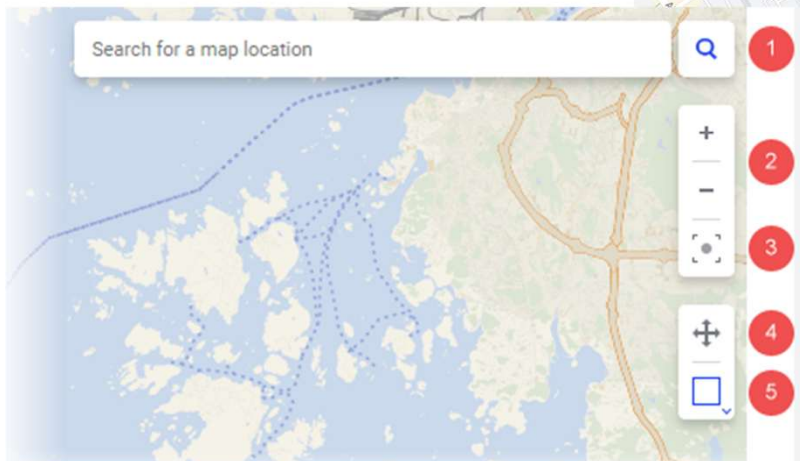
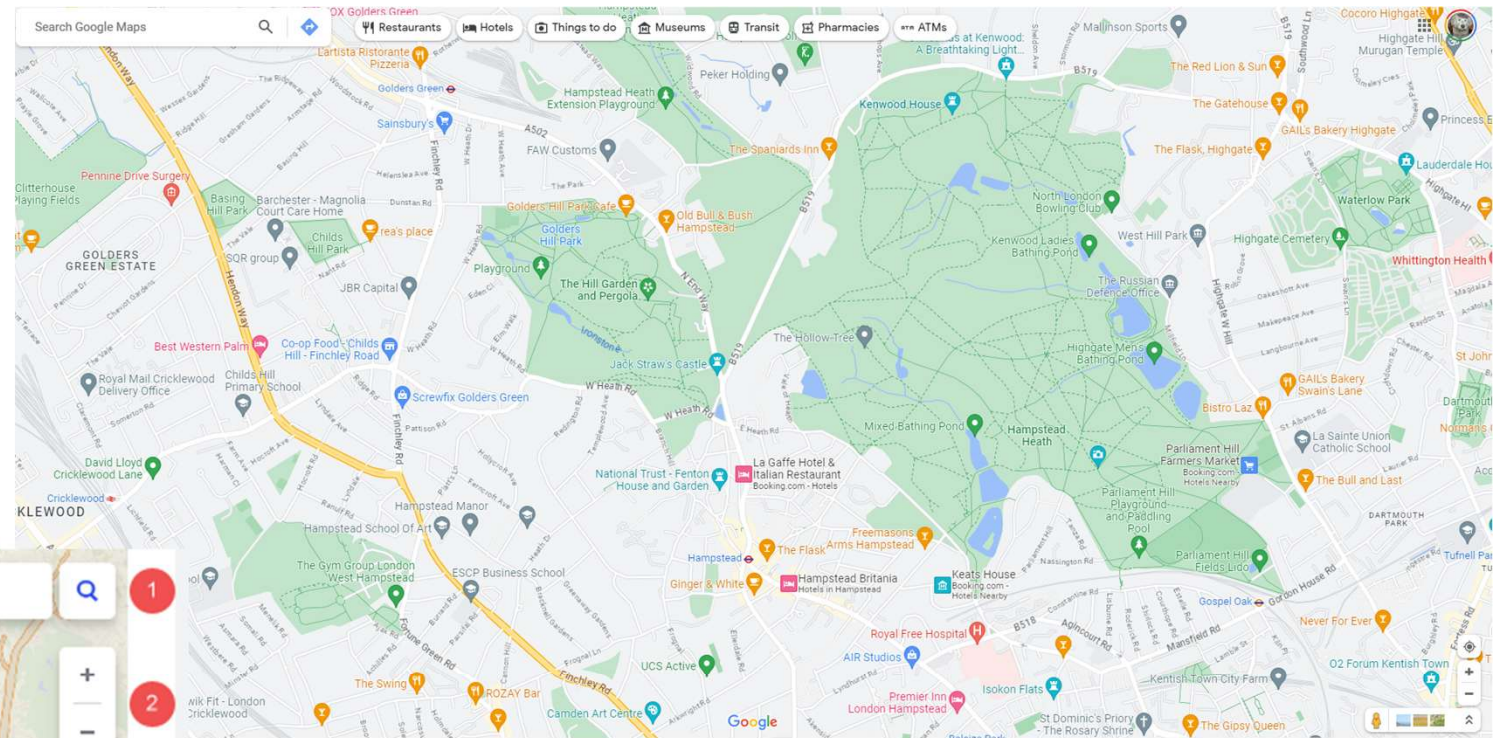


Use mouse scroll to zoom. Use click and drag to pan. Use the toolbar to zoom or reset.

Using plotly: panning script

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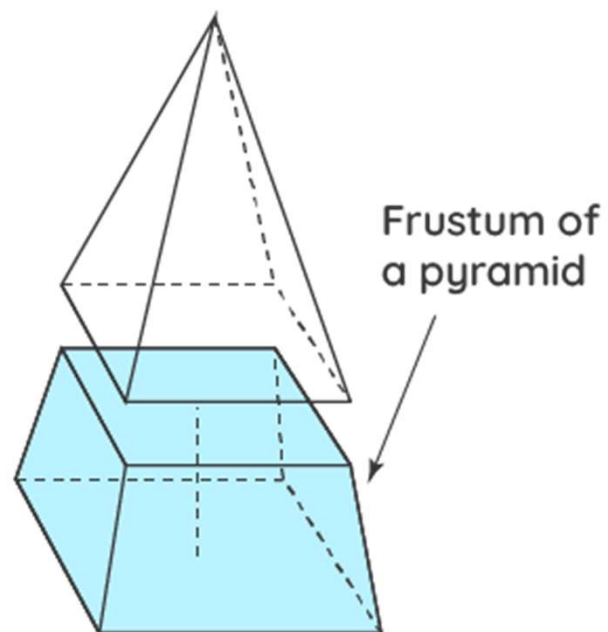
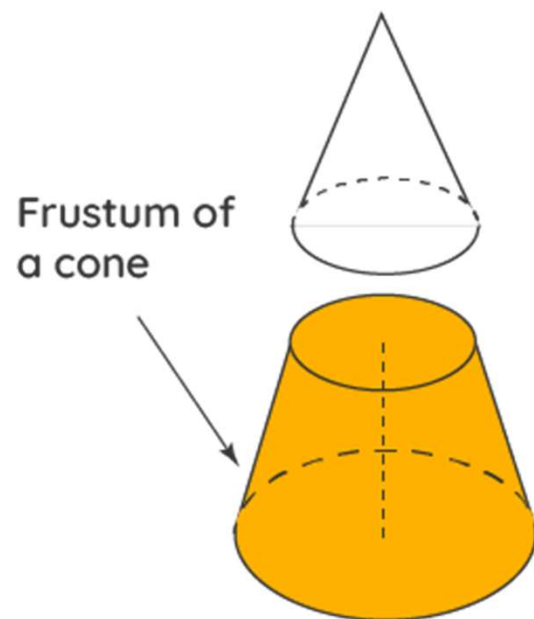
Panning



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Frustum

Types of Frustum



Frustum

...the viewing frustum represents the user's current focus of attention. Interaction techniques can leverage the frustum to:

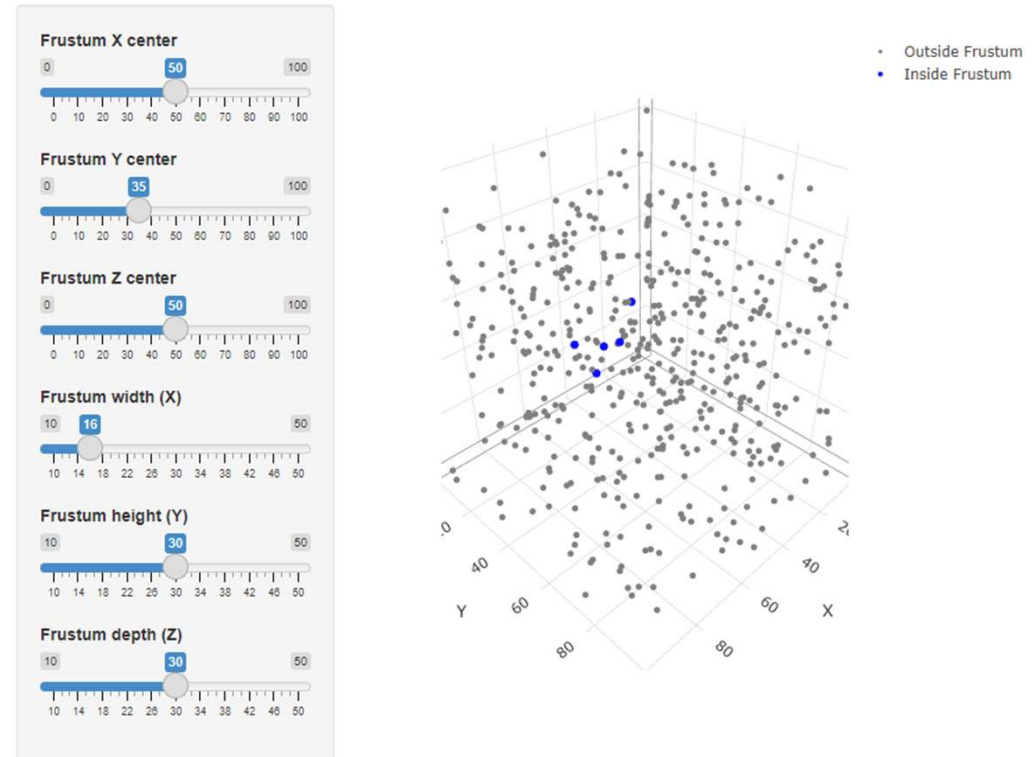
Role	How the Frustum Helps
Navigation	Determines what the user is seeing and allows interactive panning, zooming, rotating.
Filtering	Only data inside the frustum is selected or emphasized — reducing cognitive overload.
Highlighting	Brings visual or semantic focus to what is within the frustum.
Progress Tracking	Systems can track what portions of data the user has already seen (via frustum logs).

We can think of “View in Frustum” as a design pattern in interaction support, similar to “Overview+Detail” or “Brushing + Linking”.

Frustum 3D

- The **frustum** is a cuboid defined by center + width/height/depth sliders.
- Points **inside** the cuboid are rendered in **blue**.
- Points **outside** are shown in **gray**.
- The frustum can be moved through 3D space via sliders — mimicking camera-based focus.

3D Scatterplot with Simulated Frustum

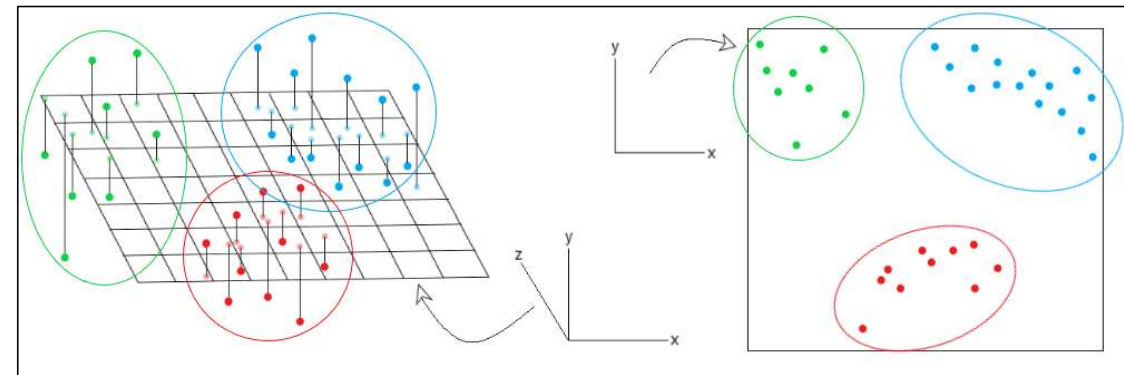
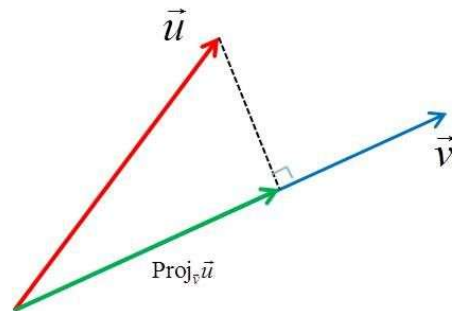


Projections

Projection

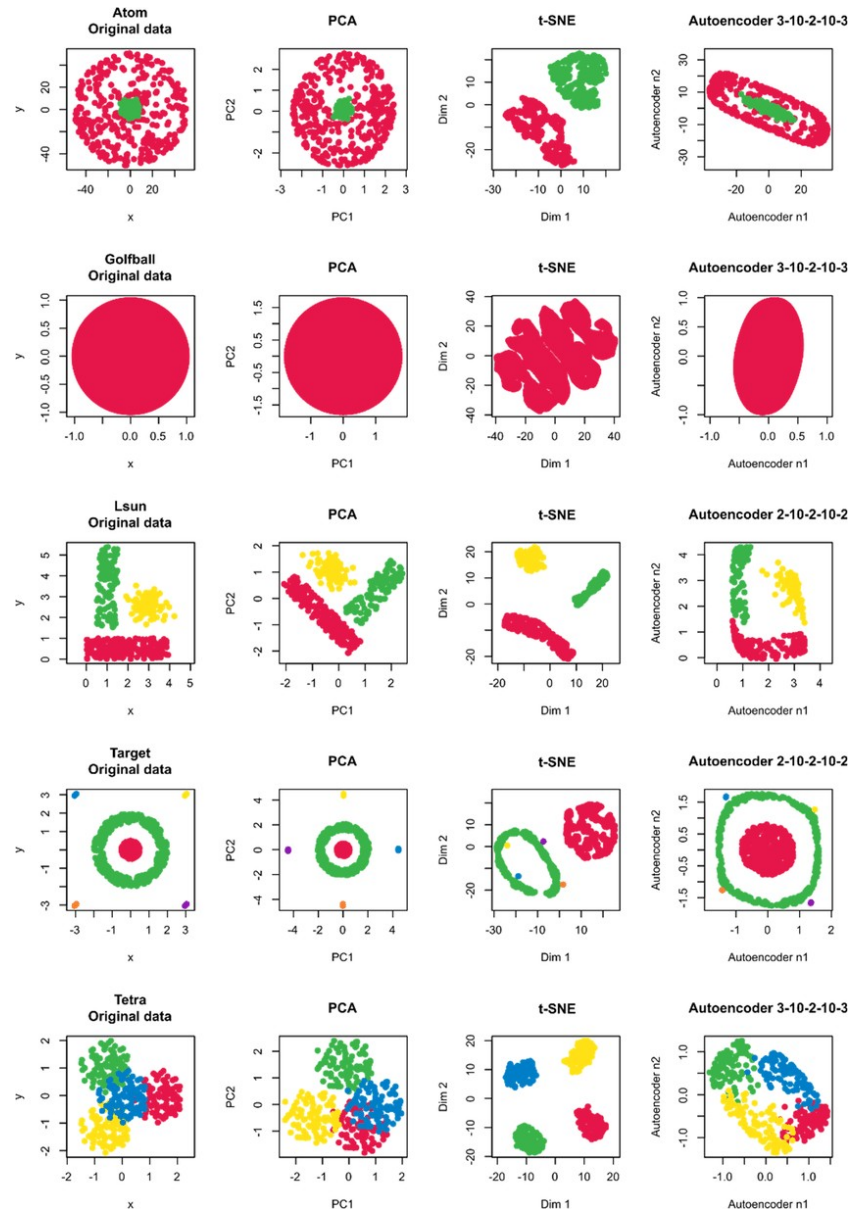
- To “project” a vector $\mathbf{u}$ onto $\mathbf{v}$ (basically, rotate a vector and place it on a second vector);

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \right) \mathbf{v}$$



Examples

Examples-of-the-
resulting-data-
projections-when-
applying-different-
projection-methods



Grand Tour (data visualization)

From Wikipedia:

- The **grand tour** is a technique originally developed by Daniel Asimov 1980–85, which is used to explore [multivariate statistical data](#) by means of an animation.
- The animation, or "movie", consists of a series of distinct views of the data as seen from different directions, displayed on a computer screen, that appear to change continuously and that get closer and closer to all possible views.
- This allows a human- or computer-based evaluation of these views, with the goal of detecting patterns that will convey useful information about the data.

Grand Tour (data visualization)

- Automated exploration method for multidimensional data.
- Allows navigation/exploration through high-dimensional data spaces by smoothly transitioning between different projections of the data onto lower-dimensional views, typically two dimensions.
- Constructs a continuous path in the high-dimensional space and then projecting the data onto two dimensions along this path.
- Users can control the speed and step size of the tour, which affects the smoothness and number of projections to be inspected.

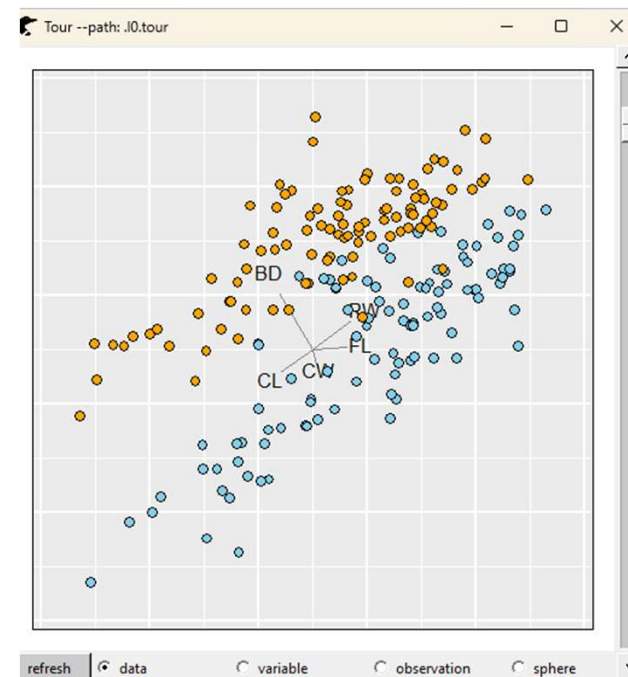
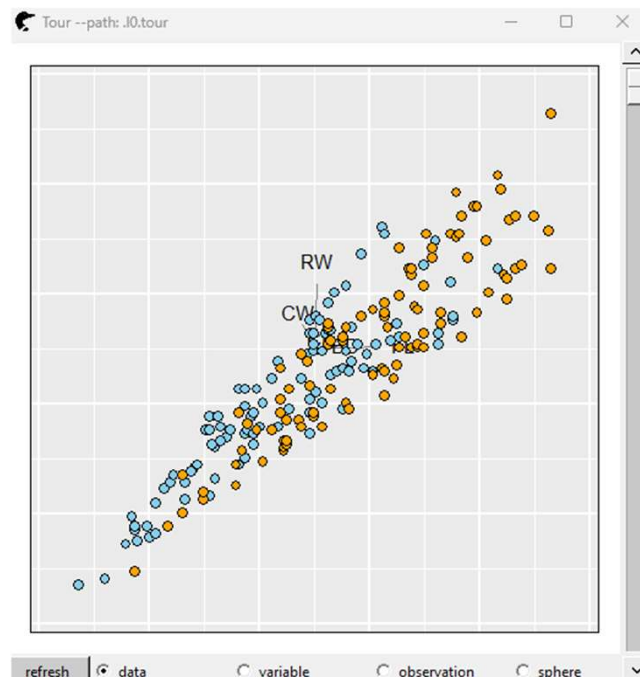
Grand Tour (data visualization)

- As the tour progresses, the user can observe how the data distributions and relationships change across different projections, gaining insights into the structure and patterns of the data.
- The grand tour is particularly useful for understanding complex, high-dimensional datasets where direct visualization becomes challenging.
- It helps in revealing hidden correlations, clusters, and patterns that may not be apparent in the original high-dimensional space. By dynamically exploring different projections
- Users can uncover interesting visual structures and anomalies, facilitating data analysis, exploration, and discovery.

Grand Tour (data visualization)

- A sequence of low-dimensional projections is created by a high dimensional data set. Tours are thus used to find interesting projections. Between each plane, interpolation along a geodesic path is provided so that the points in a sub space (e.g. 2D) can be rotated smoothly.

Grand Tour (data visualization)



The crabs (Campbell and Mahon 1974) data frame (stored in package MASS) contains 200 observations and 8 features, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes.

Projection Pursuit

From Wikipedia:

Targeted projection pursuit is a type of statistical technique used for [exploratory data analysis](#), [information visualization](#), and [feature selection](#). It allows the user to interactively explore very complex data (typically having tens to hundreds of attributes) to find features or patterns of potential interest.

Projection Pursuit

- The goal of projection pursuit is to identify projections that maximize certain desirable properties or characteristics of the data. Used to explore and analyze high-dimensional datasets.
- Focuses on finding interesting projections of the data that reveal meaningful structures, patterns, or anomalies.
- A set of candidate projections or transformations are generated and evaluated based on a chosen criterion or measure of interestingness.
- The criterion can be defined based on specific goals or insights sought from the data. Common criteria include visualizing clusters, identifying outliers, preserving certain distances or relationships, or maximizing data variance.

Projection Pursuit

- The search for interesting projections often involves an optimization process where different candidate projections are assessed and ranked based on the chosen criterion.
- This can be achieved through iterative algorithms that explore the space of possible projections, refining them to identify the most informative or visually appealing ones.
- Helps in reducing the dimensionality while preserving important structures or patterns.

Grand Tour and Projection Pursuit

Visual difference:

- Grand Tour: constantly moving, smooth transitions, exploratory.
- Projection Pursuit: jumps to interesting projections based on objective function or animates through a few optimized paths.

	Grand Tour	Projection Pursuit
Visual Philosophy	"Let the human spot interesting views"	"Let the algorithm find the best views"
Process	Continuous, exhaustive rotation	Discrete, optimized selection
Strength	General discovery	Targeted insight

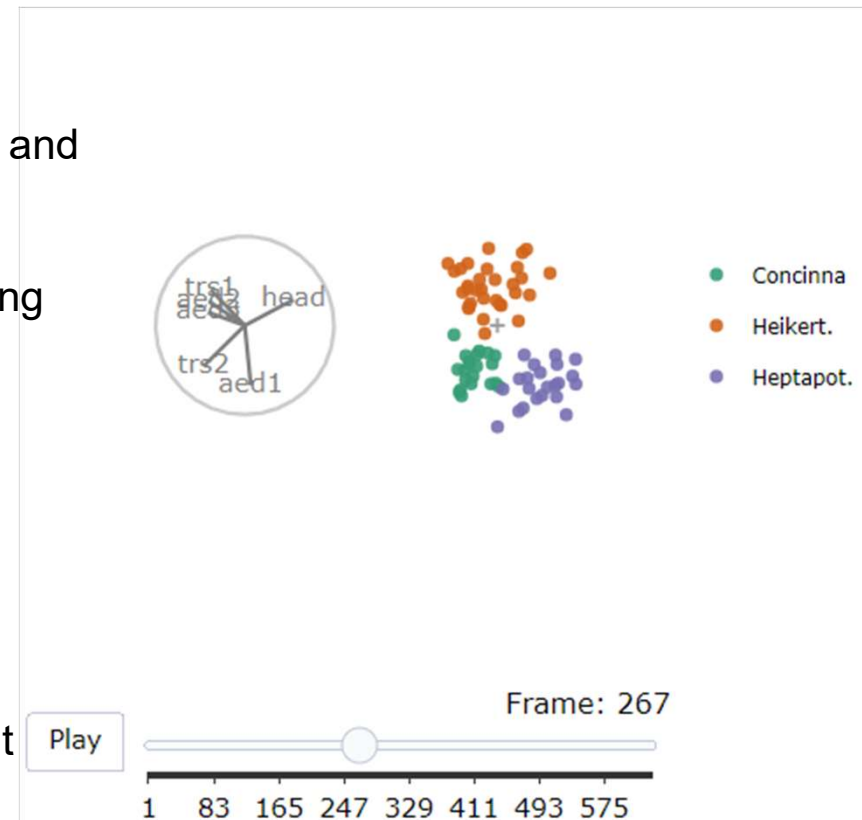
Grand Tour with Flea database

Right side: 2D Projection of Flea Data

- Each **point** represents a flea sample, originally described by 6 features (e.g., head width, segment length, etc.).
- The 6D data is being **projected onto a 2D plane** (X and Y) using the **Grand Tour**.
- The **colors** correspond to flea **species**
- These points move **as the projection changes** during the tour, revealing different "views" of the high-dimensional structure.

Left side: Projection Axes (Basis Vectors)

- **2D circle plot** showing how each original variable contributes to the current projection.
- Each arrow (e.g., aed1, trs2, head) shows how much that variable contributes to the current 2D axes.)
- The more an arrow points toward an axis, the more it contributes to the projection in that direction.

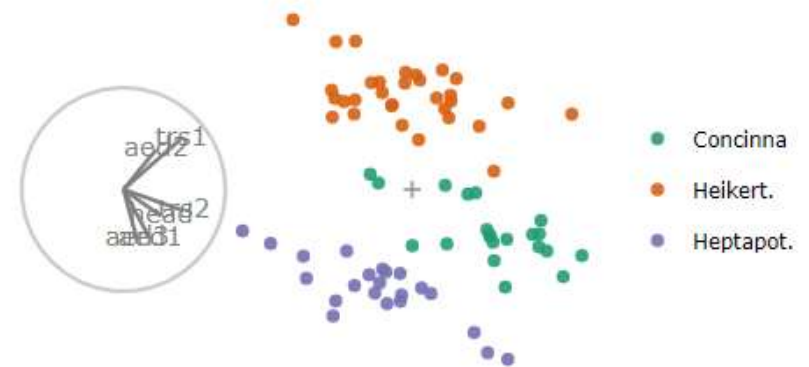


Projection Pursuit with Flea database

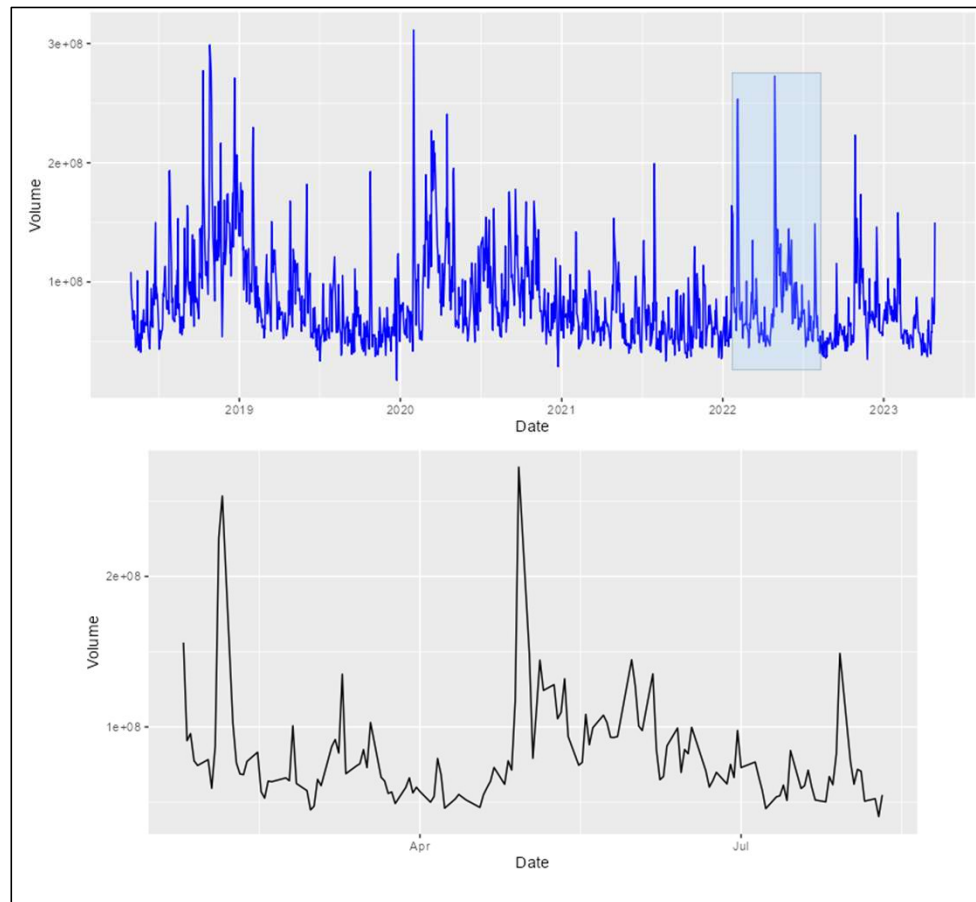
This is a **Guided Tour** using **Projection Pursuit with LDA** as the optimization target.

The image shows:

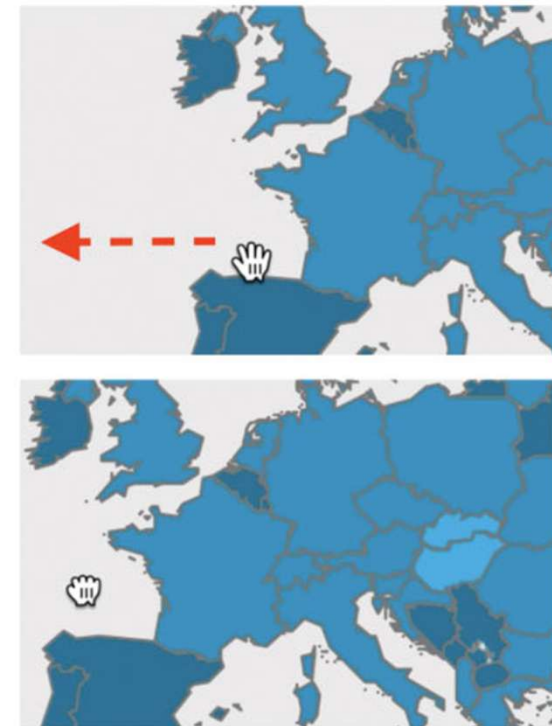
- A **single projection** (i.e., a single 2D view).
- Points are very **well-separated by class** (much more than in the grand tour), because the tour **found a projection that maximizes class separation**.
- The **arrows** (circle plot on the left) indicate how each original variable contributes to this optimal projection.
- Unlike the grand tour, the **clusters remain mostly fixed**, because this view has already been optimized for maximum class separation.



Interaction Operators – exploration examples



(B) Explore



Using Shiny R

6/13/2026 Source: <https://mastering-shiny.org/action-graphics.html>

Interaction Operators - reconfigure

- Reconfigure refers to the manipulation and rearrangement of data and visual representations to provide different perspectives and highlight specific features or patterns.
- By reorganizing and transforming the data, users can gain new insights and uncover hidden characteristics of the dataset.

Interaction Operators - reconfigure

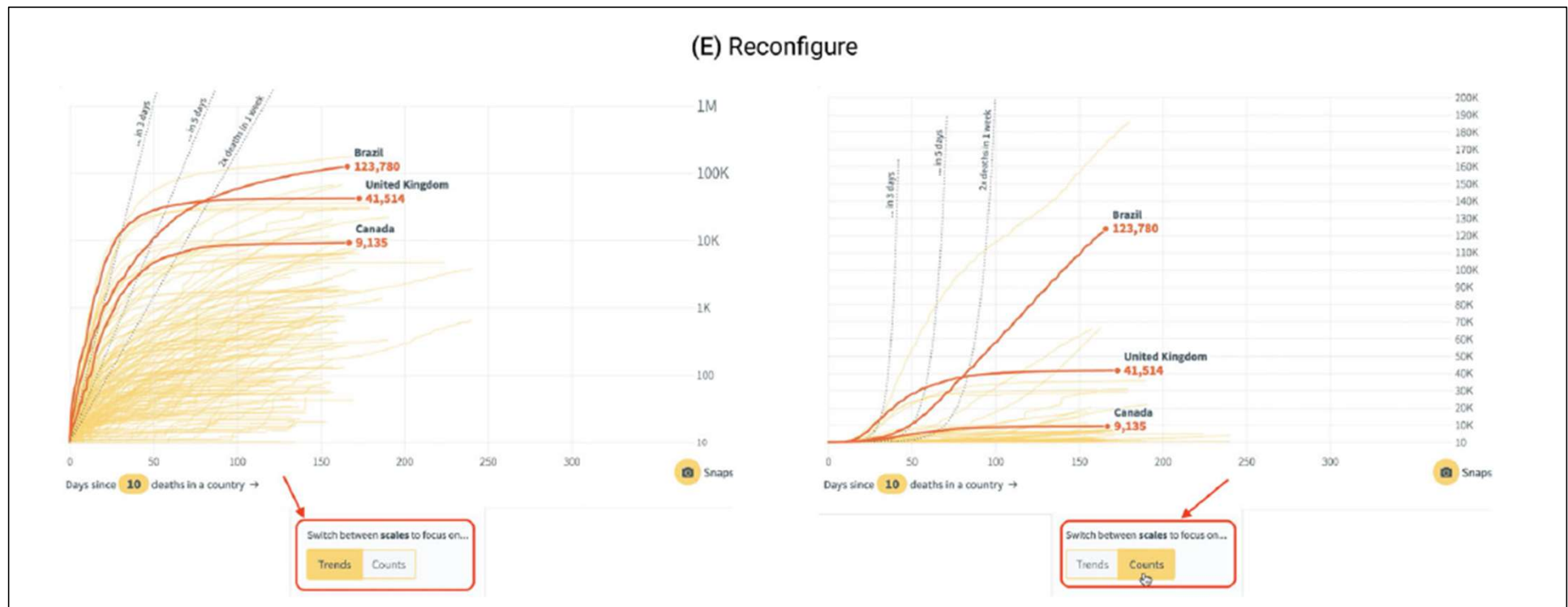
- Reconfigure interaction techniques can enable users to:
 - change the spatial arrangement of visual representations to provide different perspectives on the data.
 - Examples of reconfigure techniques include sorting and rearranging columns in table-based visualizations, changing the attributes assigned to the axes in scatter plots, and baseline adjustment in stacked histograms. These techniques allow users to explore relationships, compare attributes, and better understand the data.
 - reduce occlusion in complex visualizations where data points may overlap or occlude each other, interaction techniques such as view rotation in 3D visualizations or applying jitter to the position of data items can help reduce occlusion and reveal more information.
 - freely move and arrange data items to suit their mental model or create custom groupings. Examples include freely moving nodes in a social network visualization or arranging web browser favorites on an inclined plane.

Interaction Operators – reconfigure example



Source: Appendix: The interaction techniques [1] instantiated in each visualization
part of the following paper: Fan, M., Wang, Y., Xie, Y., Li, F. M., & Chen, C. (2022). Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think-Aloud Protocol.
International Journal of Human-Computer Interaction

Interaction Operators – reconfigure example



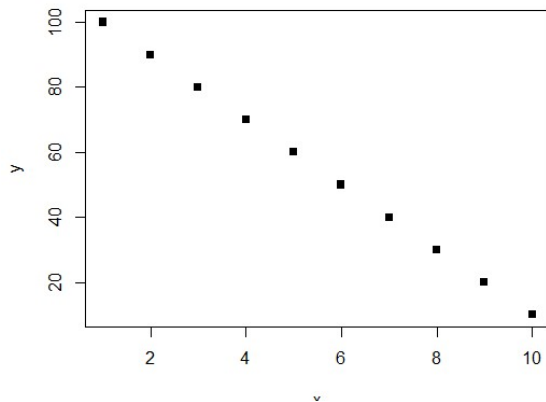
Source: Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think-Aloud Protocol, Mingming Fan, 26 May 2022

Scatterplot Jittering - example of reconfiguration

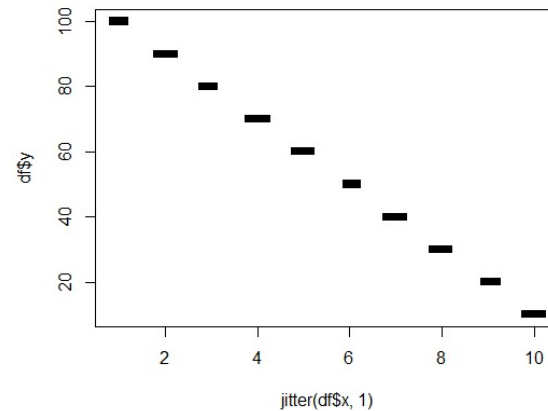
- Sometimes we have discrete data that is almost continuous (e.g., years of formal education). These kinds of variables might be nearly continuous and have approximately linear relationships with other variables.
- We need to preprocess the data in order to clearly visualize it.
- Add noise and run a regression model on the original data then when we plot the results use the jitter inputs in order to more clearly convey the underlying descriptive relationship.
- Also helps when many data points have the same coordinates, and we want to see them.

Scatterplot Jittering

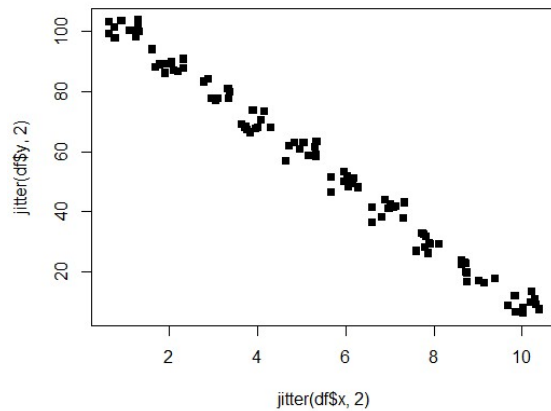
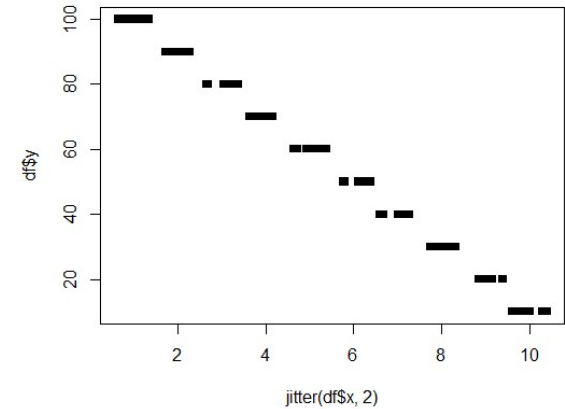
`plot(df$y ~ df$x, pch = 15)`



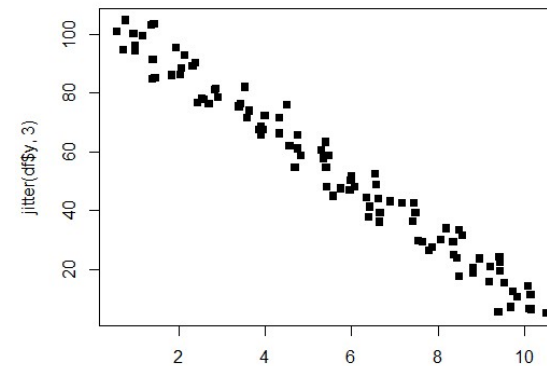
`plot(df$y ~ jitter(df$x, 1), pch = 15)`



`plot(df$y ~ jitter(df$x, 2), pch = 15)`



`plot(jitter(df$y, 2) ~ jitter(df$x, 2), pch = 15)`



`plot(jitter(df$y, 3) ~ jitter(df$x, 3), pch = 15)`

Scatterplot Jittering

- In the context of interaction operators, **jittering is best classified as Reconfigure**, since it rearranges the **visual layout** of data items to reduce occlusion.
- While it could also be seen as an **encoding adjustment**, the key distinction is that jittering **repositions elements** rather than changing how their values are depicted.
- It may also support **elaboration** by helping users access hidden or overlapping data, but its primary role is to reconfigure spatial relationships for clarity.

Interaction Operators - encode

- Encode refers to the manipulation of visual representations/attributes and the visual encoding of data values.
- Change visual appearance, such as font, color, size, shape, or orientation, of data elements to adapt the visual encoding to suit users needs and gain new insights from the data.
- These techniques can be used simultaneously and combined to encode multiple variables into a visual representation.

Interaction Operators - encode

- Adjust color encoding dynamically, experiment with various color schemes, and understand distributions of multiple variables.
- Different visual encodings can reveal features of the data that may be difficult to perceive with a single mapping.
- Examples:
 - overlapping points in a scatterplot representing years can be represented uniquely in a parallel coordinate display.

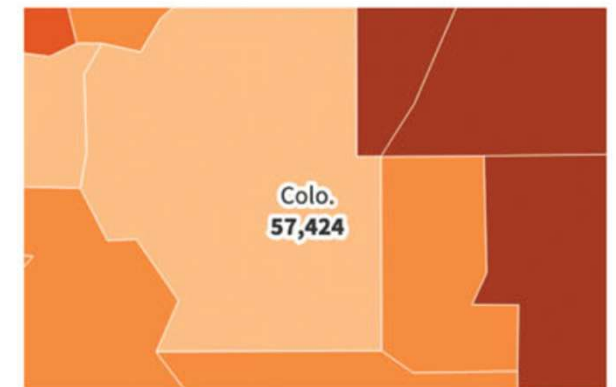
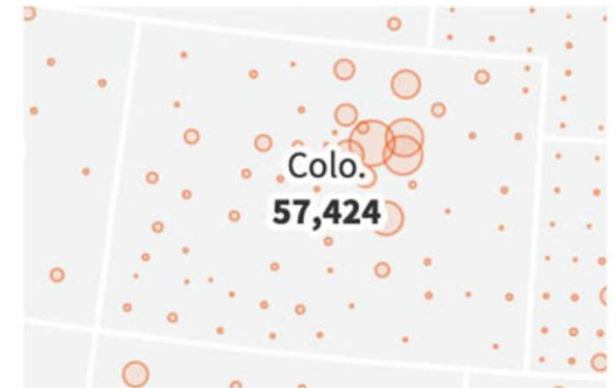
Interaction Operators - encode

Visual encoding techniques allow users to:

- Visual elements not only affect pre-attentive cognition but also play a crucial role in understanding relationships and data distributions.
- Change the representation type, such as transforming a pie chart into a histogram to uncover new aspects of relationships.

Source: Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think-Aloud Protocol, Mingming Fan, 26 May 2022

(C) Encode



Interaction Operators - Abstraction/Elaboration

- Abstract/elaborate involve adjusting the level of detail and abstraction in a data representation, allowing users to switch between broad contextual views and detailed examinations of individual data cases.
- Hierarchical structures of time/geography/organization, with higher-level abstractions providing aggregated overviews and lower levels containing more detailed information, are well-suited for interactive drill-down exploration in time-oriented data.

Interaction Operators - Abstraction/Elaboration

- Zooming, whether geometric or semantic, allows users to see an overview of a larger dataset (zoom-out) or a detailed view of a smaller dataset (zoom-in) while maintaining the representation's integrity.
- Zooming brings details into focus or provides context as they fade away.

Distortion

- Distortion operators, often used interactively, enable users to focus on a subset of the data while reducing detail in other parts.
- Distortions can be applied to any type of visualization and vary in their effects and preservation of context.
- Methods that distort the entire space being analyzed and others that have more localized effects.
- May take place within the original visualization or may appear in a separate window

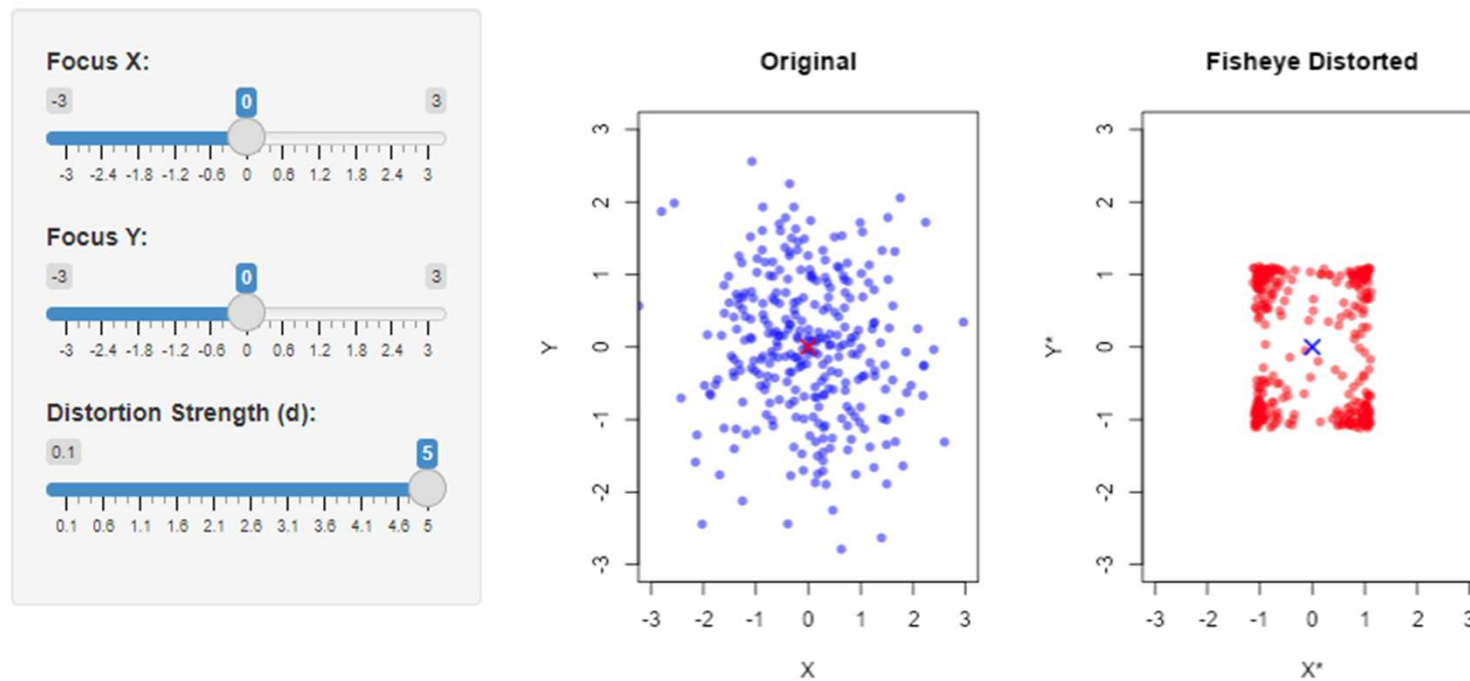
Distortion

- Vary in the features that are preserved and the amount of context maintained.
- Example: text distortion techniques strive for readability within a small region of interest, with the rest of the text positioned to reinforce document structure, but not generally readable.
- Other types of distortion: it is important that the undistorted and compressed regions continue to convey useful information while details are provided in the focus area.
- Distortion operators may be linear or non-linear

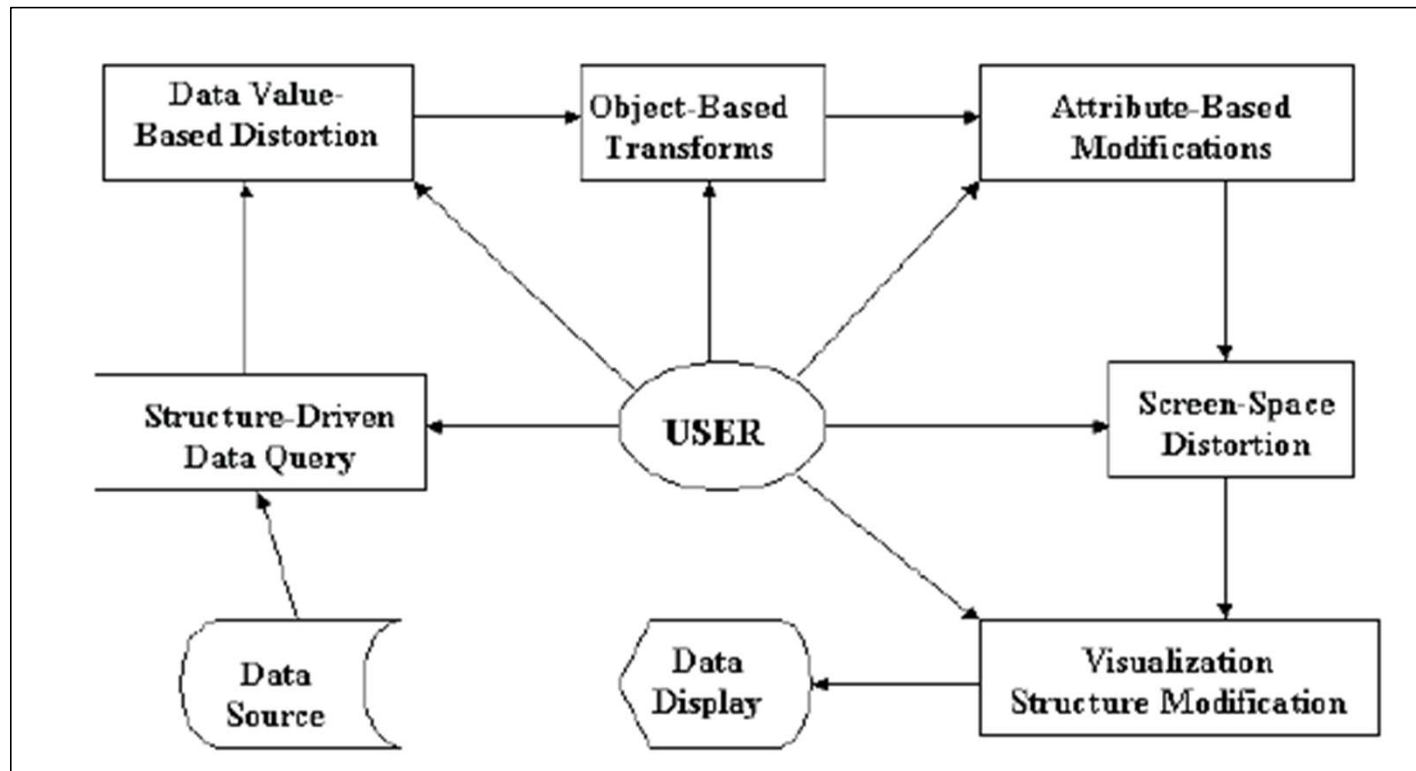
Distortion

- Move the focus point to any position using the sliders.
- Change the distortion level (d) to increase or decrease the fisheye effect.
- See side-by-side plots: original vs distorted.

Interactive Fisheye Distortion



The distortion pipeline



The user interactively controls each stage of the pipeline.
Each distortion operation is optional. Source: Ward page 383

Interaction Operators - Abstraction/Elaboration (example)



6/13/2026

Using Shiny R

Source: <https://neuralearn.net/how-to-create-drilldown-graphs-with-plotly-in-r/>

Source: Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think63 Aloud Protocol, Mingming Fan, 26 May 2022

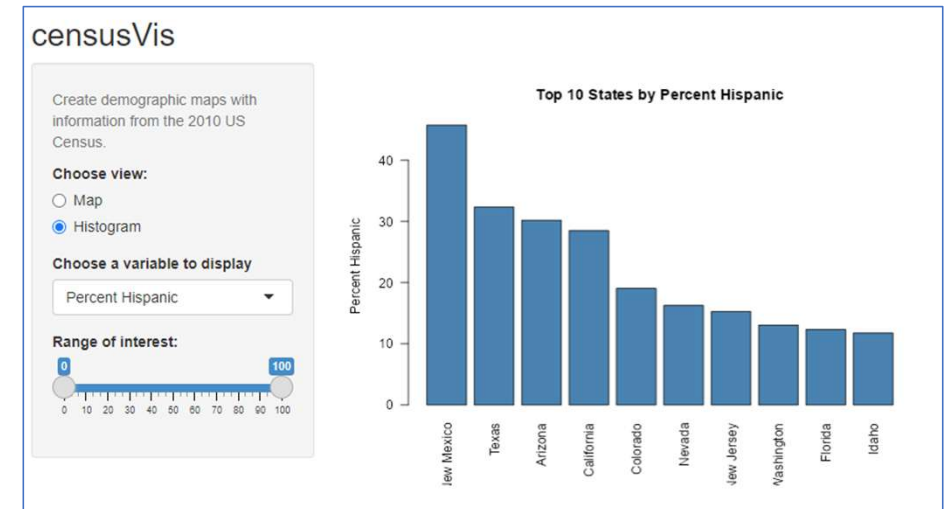
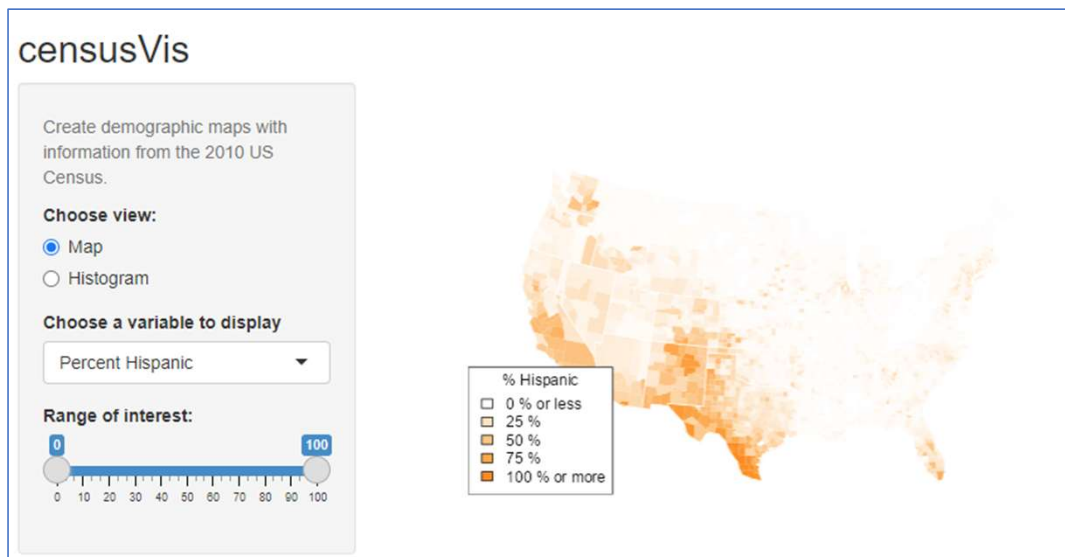
Interaction Operators - Filter

- Allows users to reduce the volume of data to be visualized by setting constraints to specify the data to be preserved or removed.
- Often indirectly through a separate interface or dialog box, prior to viewing the data, to avoid overloading the display.
- Filtering can be done:
 - with dynamic query specification, using sliders or other controls, allows users to specify a range of interest, and the visualization is immediately updated to reflect the changes.
 - by selecting items from a set or list to preserve or hide. Filtering simplifies the view and eases interpretation of the visualization.
 - directly by indicating objects on the visualization. The operation performed on the selected subset can be indistinguishable from a filtering operation.

Interaction Operators - Filter

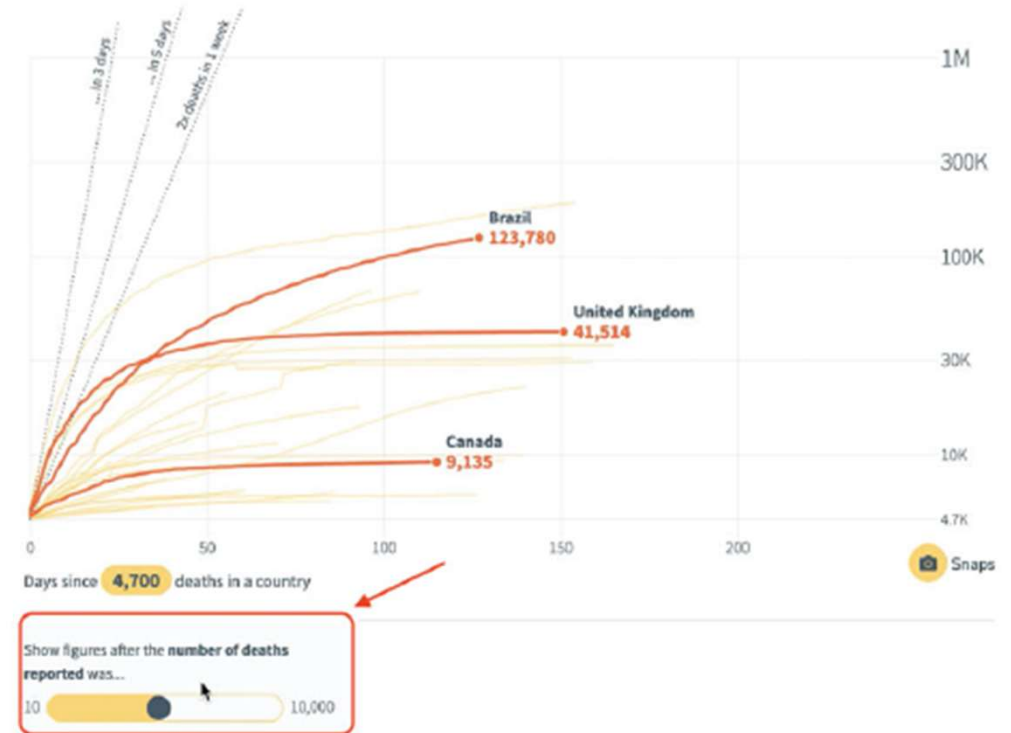
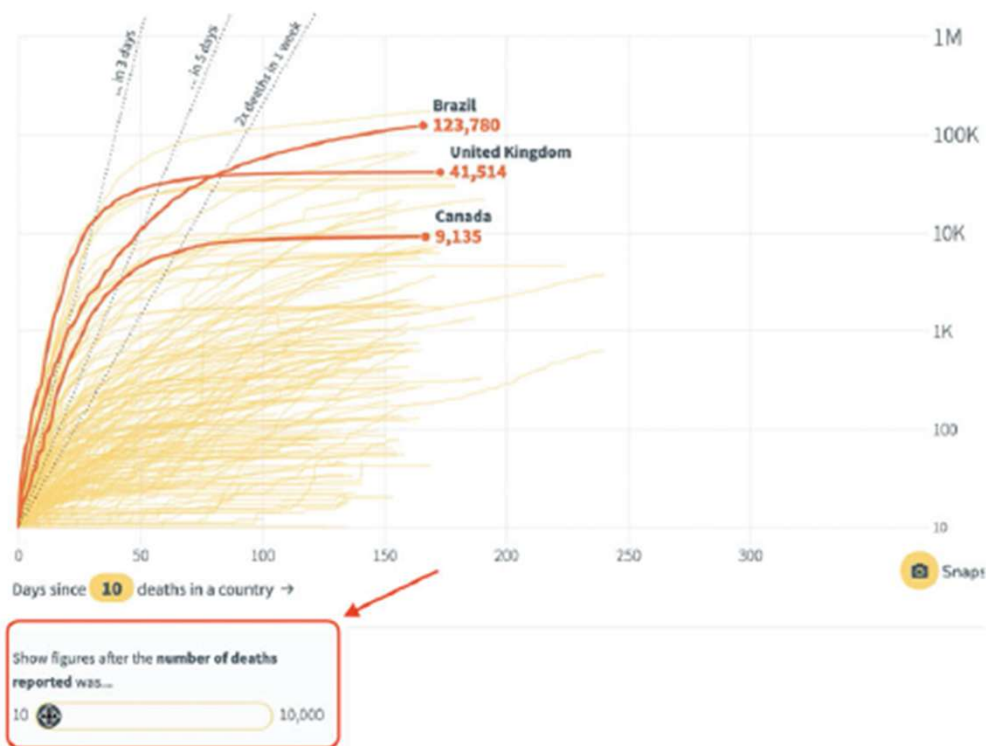
- The actual data usually remain unchanged so that hidden or differently shown data items can be recovered.
- Some systems extend dynamic query capabilities by changing the colors of filtered data items instead of removing them from the display. This helps users understand the context of the dataset by showing nearby data items that do not fully meet the filtering criteria.

Interaction Operators - filtering example



Filtering

(F) Filter



Source: Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think-Aloud Protocol, Mingming Fan, 26 May 2022

Interaction Operators - Connect

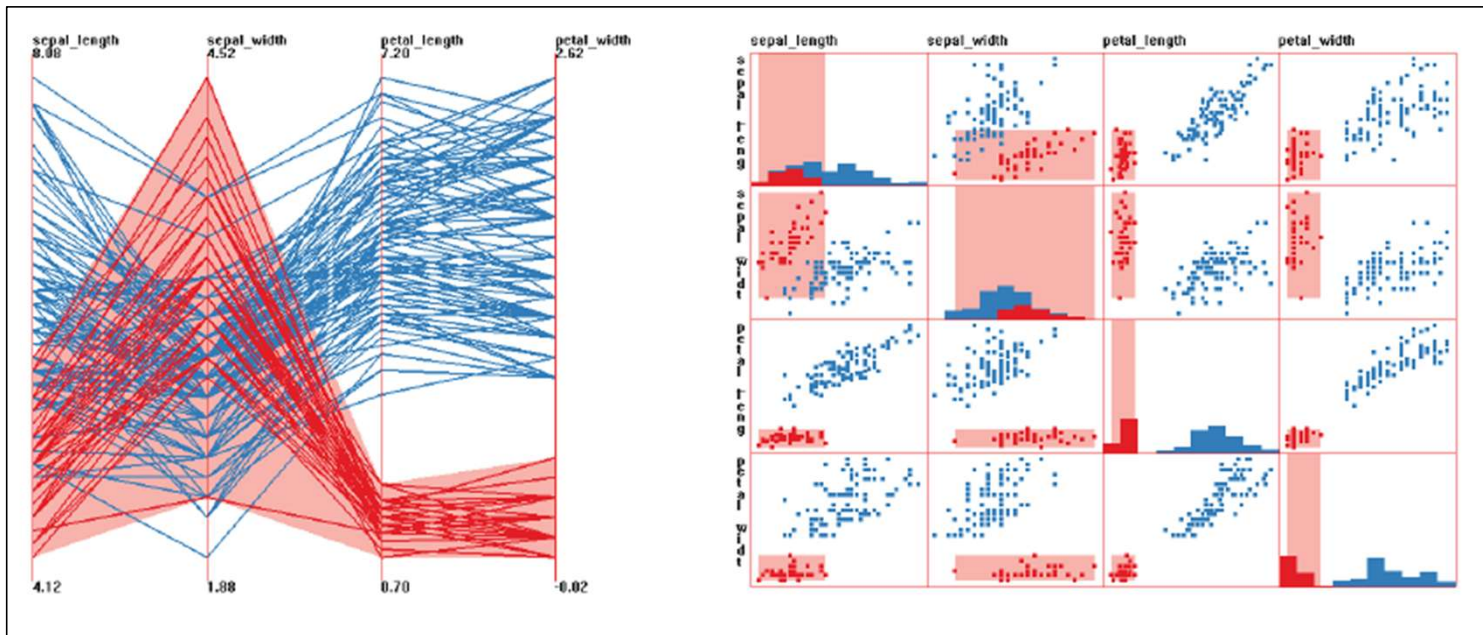
- Connect (link) establishes links between selected data in one view and the corresponding data in other views.
- Used to reveal interesting features and relationships in the data.

Interaction Operators - Connect

Linked selection/brushing allows users to:

- (selection) highlight a feature in one view and gain a more complete understanding of that feature by seeing how it appears in other views. Linked brushing provides dynamic displays that show changes in values across linked displays.
- (brushing) specifying complex constraints on selections. Different types of views are optimized for conveying specific types of information and specifying conditions with accuracy. By combining multiple views, users can specify different aspects of a query effectively and intuitively.
- Helps in building a mental model and discovering relationships between different attributes of the data set. Can be applied to multiple views where brushing is used to highlight selected data items across views.

Interaction Operators - connect example



Source: Interactive Data Visualization, Foundations, Techniques, and Applications - Ward page 371

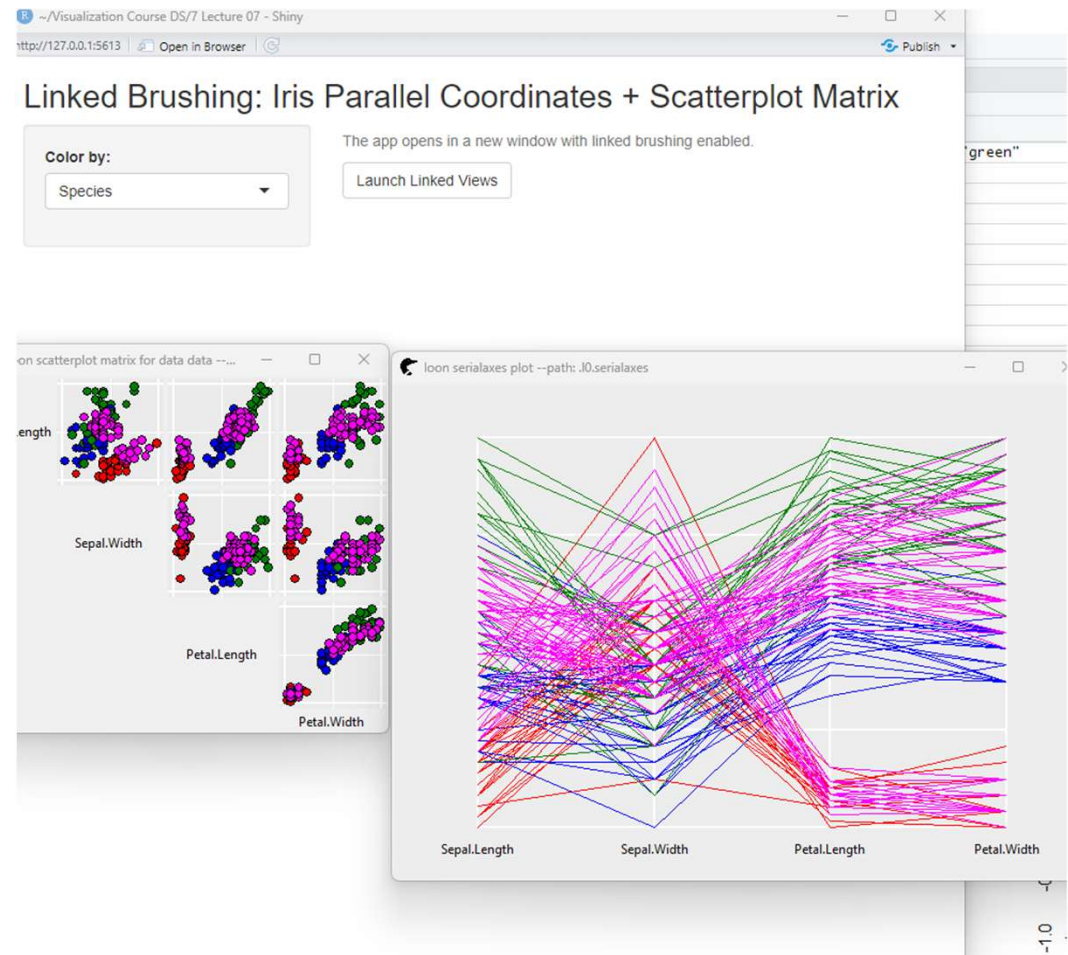
Example of linked brushing. A cluster is isolated in parallel coordinates, with linked selection in the scatterplot matrix. Selected data is dark red, while brush extents are shown as light bands or rectangles.

Linked Brushing using Shiny R

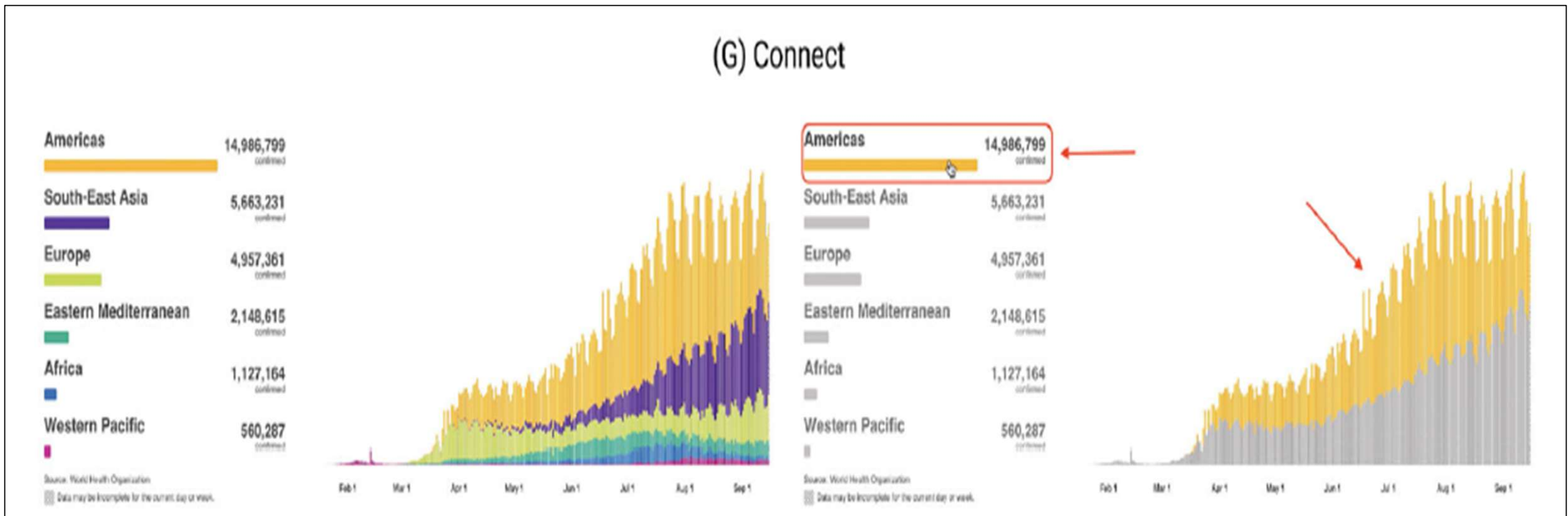
- A **parallel coordinates plot**
- A **scatterplot matrix**
- Can **brush** in either — and both will update.

In magenta the selected data points.

Shiny linked brushing



Interaction Operators - connect example



Source: Understanding How Older Adults Comprehend COVID-19 Interactive Visualizations via Think-Aloud Protocol, Mingming Fan, 26 May 2022

Interaction Operands and Spaces

- To determine the result of an interactive operation, one needs to know within **what space** the interaction is to take place.
- Or: when a user clicks on a location or set of locations on the screen, what entities does he or she wish to indicate?
- An interaction operand is the section of space upon which an interactive operator is applied.
- Possibilities include the:
 - pixel(s) → applied to the screen
 - data value or record mapped to the location
 - component of the visualization structure (e.g., an axis) at or near that location.

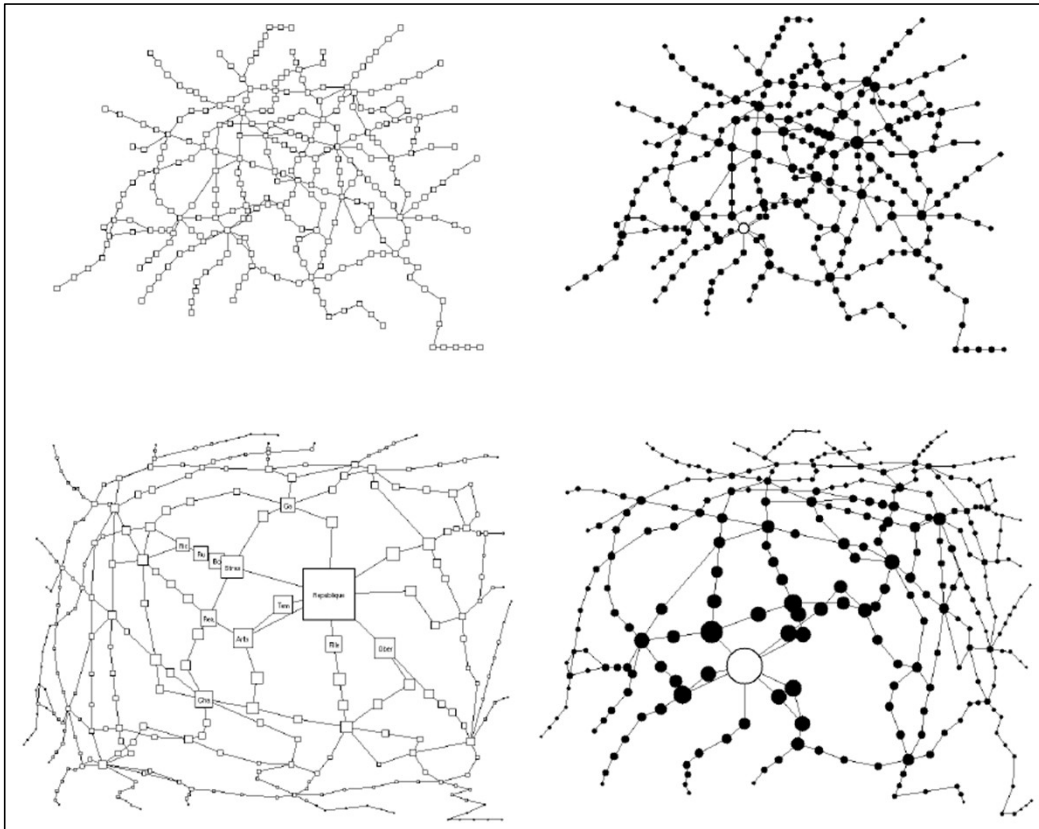
Screen Space (Pixels)

- Screen space consists of the pixels of the display.
- Navigation in screen space typically consists of actions such as panning, zooming, and rotation.
- No new data is used; the process consists of pixel-level operations such as transformation, sampling, and replication.

Screen Space (Pixels)

- Pixel-based selection means that at the end of the operation, each pixel will be classified as either selected or unselected.
- Examples of screen space techniques are the fisheye lens and rubber sheet methods.
- Suitable Datatypes : Graphs, particularly network topologies, VLSI circuits and graph theory.

Screen Space (Pixels) - example



Images represent the Metro system in Paris, whereat the upper left image is a normal view of the Metro.

In the lower left image, the representation of the vertices is changed so that the names of the stations are shown, space permitting. [Sarkar and Brown, 1993]

Source: <https://infovis-wiki.net/wiki/File:FisheyeView.png>

Data Value Space (Multivariate Data Values)

- Operations are applied directly to the data, involve using the data values as a mechanism for view specification.
- Analogous operations for:
 - panning and zooming would be to change the data values being displayed;
 - panning would shift the start of the value range to be shown,
 - zooming would decrease the size of this range.
 - filtering could include additional transformation of the remaining data.

Data Value Space (Multivariate Data Values)

- Data value space selection is like a database query in that the user specifies a range of data values for one or more data dimensions.
 - Can be performed via direct manipulation, as in the data-driven brushing reported in or via sliders or other query specification mechanisms.
 - Selection may involve a single value, or one or more ranges of values.

Data Value Space (Multivariate Data Values)

- For extremely large data sets, first reduce the data to a particular region of data space.
 - Sampling might be used to examine a representative subset of a large data repository where the visualization might otherwise be too cluttered to discern any useful patterns.
- Filter dimensions to allow examination of a subset of dimensions with similar characteristics or select representatives from clusters of related dimensions.
- One of the most powerful reconfiguring operators is in reordering (e.g., sorting) the records and/or the dimensions.
- Different ordering functions will expose different subsets of patterns within the data.

Data Structure Space (Components of Data Organization)

- Selection in data structure space involves displaying the structure and allowing the user to identify regions of interest within it.
- Filtering is often performed in data structure space to reduce the amount of information on the display.
 - Time-series visualization, it is common to identify a range in the time axis (implied by the data ordering) on which to focus one's attention.
 - Social networks: filtering out nodes and links that are greater than a particular number of links away from a focus point.
- Abstraction/elaboration operators take many forms in data structure space.
- In hierarchical structure:
 - abstraction simply means moving up the hierarchy to aggregations or summarizations of the data at the current level,
 - elaboration implies drilling down into the hierarchy.

Attribute Space (Components of Graphical Entities)

- Operators are focused on one or more of the attributes associated with the graphical entity being used to convey information.
- Attributes could include color, size, shape, or any other of the eight visual variables.
- For example, distortion of a color map would allocate a wider or narrower range of colors for some subranges than for others, thus enabling fine variations to be more readily perceived
- This form of distortion is often used in medical image analysis to identify regions of interest.

Interaction Techniques – spaces (summary)

- Screen Space
- Data Space (Multivariate Data Values)
- Data Structure Space (Components of Data Organization)
- Visualization Structure Space (Components of the Data Visualization)
- Visual Attributes

Interaction Techniques - Data Space (Multivariate Data Values - distortion)

- Scaling and translating to fit a particular range of values
- Exponential or log scaling to spread or compress the distribution of the data
- Sinusoidal functions to help study cyclic behavior
- Absolute value transformation to focus on magnitudes of change
- Negating the values so that low values become high, and vice versa

Interaction Techniques - Data Space - key requirements

- User be made aware that a transformation on the data has occurred.
- Clear labeling of axes is essential: augmenting the visualization with a note describing data transforms that have been applied to reduce misinterpretation.

Interaction Techniques - Control

- Users require mechanisms to control the type, location, and level of each interaction to navigate within both the data and the visualization.
- Controls must be intuitive, unambiguous, and at a level of detail and accuracy appropriate for the space being operated upon.
- Extension Control: It may be specified via a single value (e.g., a radius or maximum number of items) or via a vector of values (e.g., a range for each data dimension or a set of constraints).
- In many systems, the extents are often hard-coded to reduce the effort in performing the operation.
- Interaction type selection Icons or buttons in the user interface could also be used, as the number of choices would normally be modest.

Appendix

Unified Framework

For each interaction operator to be applied to a specified space/operand, several parameters are required:

- Focus: The location within the space at the center of the area of user interest.
- Extents: The range within the space defining the boundaries of the interaction.
- Transformation: The function applied to the entities within the extents, generally a function of distance or offset from the focus. The shape of this transformation might also depend on the type of information being affected.
- Blender: how to handle parts of space touched by more than one interaction. For selection, this operation may include performing union, intersection, or other logical operations on overlapping entities .
- For distortion, there are several approaches, including weighted average, maximal value, and composition.

Interaction Techniques - Control

- **Focus selection:** direct manipulation tools, e.g., using a mouse or other selection device to indicate the focus location.
- **Extent selection:** depends on the type of interaction and the space in which the interaction is being applied and can be done either via direct manipulation or separate interface tools.
- **Interaction type selection:** pair of menus: one to select the space, and the other to specify the general class of the interaction.

Interaction Techniques - Control

- **Interaction level selection:** degree of interaction is an important control parameter that can be specified by a single value (e.g., the magnitude of scaling that will occur at the focal point).
- **Blender type selection:** when more than one interaction can be simultaneously viewed and manipulated, there must be some mechanism for selecting a strategy for mixing regions of space affected by more than one interaction. As with interaction-type selection, this is best accomplished via a menu of options.